Artificial Intelligence Review (2024) 57:342
https://doi.org/10.1007/s10462-024-11000-0

A survey of recent approaches to form understanding in scanned documents

Abdelrahman Abdallah[1] · Daniel Eberharter[1] · Zoe Pfister[1] · Adam Jatowt[1]

Accepted: 4 October 2024 / Published online: 21 October 2024

Abstract

This paper presents a comprehensive survey of over 100 research works on the topic of form understanding in the context of scanned documents. We delve into recent advancements and breakthroughs in the field, with particular focus on transformer-based models, which have been shown to improve performance in form understanding tasks by up to 25% in accuracy compared to traditional methods. Our research methodology involves an in-depth analysis of popular documents and trends over the last decade, including 15 state-of-the-art models and 10 benchmark datasets. By examining these works, we offer novel insights into the evolution of this domain. Specifically, we highlight how transformers have revolutionized form-understanding techniques by enhancing the ability to process noisy scanned documents with significant improvements in OCR accuracy. Furthermore, we present an overview of the most relevant datasets, such as FUNSD, CORD, and SROIE, which serve as benchmarks for evaluating the performance of the models. By comparing the capabilities of these models and reporting an average improvement of 10–15% in key form extraction tasks, we aim to provide researchers and practitioners with useful guidance in selecting the most suitable solutions for their form understanding applications.

Keywords Form understanding · Document layout · OCR · Financial document · Transformer

1 Introduction

Digitization has become ubiquitous in various aspects of our lives, where physical objects like music discs and analogue documents are being replaced with digital, machine-readable counterparts. This transformation enables downstream applications and historical preservation through efficient storage in databases (Davis et al. 2019; Zhong et al. 2019). Document

✉ Abdelrahman Abdallah
Abdelrahman.Abdallah@uibk.ac.at

[1] Department of Computer Science, University of Innsbruck, 6020 Innsbruck, Tirol, Austria

understanding (Kim et al. 2022; Abdallah et al. 2024, 2023; Luo et al. 2024; Appalaraju et al. 2024; Liu et al. 2024; Wang et al. 2024; Tanaka et al. 2024) is the field of analyzing the content and structure of documents with various formats and modalities, such as text, images, tables, and graphs. Language models and transformers (Irie et al. 2019) are advanced neural network architectures that have demonstrated remarkable capabilities in various natural language processing tasks, as well as computer vision and audio processing tasks. The transformative potential of these models in document understanding has been widely recognised (Wang et al. 2019).

Form understanding is essentially a sequence-labeling task akin to named entity recognition (NER), often referred to as *key information extraction* (KIE). The task poses unique challenges due to the diverse shapes and formats in which these documents appear (see Fig. 1 for examples). Unlike traditional NER tasks that deal with 1D text information, form understanding involves multiple modalities. In addition to textual information, the position and layout of text segments play a vital role in semantic interpretation. An effective language model for form understanding must comprehend document entities within the context of multiple modalities and adapt to various types of document formatting. One of the challenges in form understanding is the integration of visual information from scanned documents. Vision-language models, such as Vision Transformer (ViT) (Han et al. 2023; Wu et al. 2021) and Contrastive Language-Image Pretraining (CLIP) (Li et al. 2021) have demonstrated the potential to bridge the gap between text and images. These models enable the understanding of visual elements within the context of textual content, offering new avenues for improved form understanding.

A notable example of the fusion of language models and transformers in document understanding is the LayoutLM model (Xu et al. 2020). This model extends BERT by incorporating layout information, enabling it to consider the spatial arrangement of tokens in addition to their textual context. LayoutLM demonstrated superior performance in tasks like document image classification and form understanding, where both textual and spatial information are crucial. Form understanding is however still a challenging and important problem in the field of document analysis and understanding. While there are several surveys that cover general aspects of document understanding and layout analysis (Binmakhashen and Mahmoud 2019; Subramani et al. 2020), as well as specific topics such as visual document analysis (Igorevna et al. 2022), camera-based analysis of text and documents (Liang et al. 2005), and deep learning for document image understanding (Shigarov 2023), there is currently no comprehensive survey focusing specifically on the task of *form understand-*

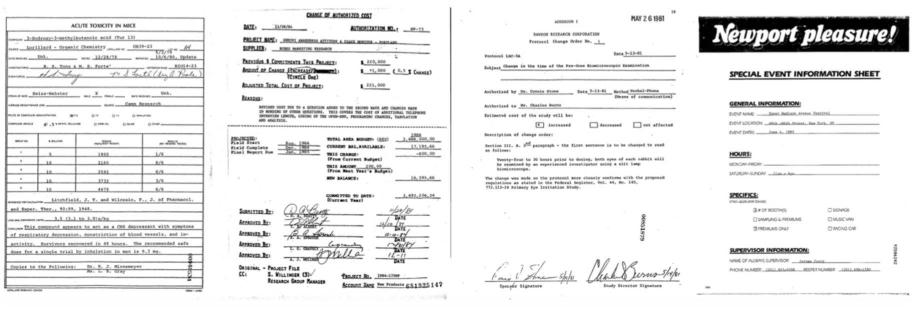


Fig. 1 Examples of scanned documents with different layouts (Xu et al. 2020)

ing. Our survey bridges this gap and provides an extensive overview of recent technologies dedicated to the form understanding task.

Among the crucial tasks in digitization, extracting form information from scanned documents stands out as a critical challenge. Digitizing form documents is non-trivial due to their inherent complexity, including intricate structures, non-textual elements like graphics, and combinations of hand- and machine-written content (Lewis et al. 2006). Moreover, scanned documents may suffer from noise, resulting in unclear or distorted images, further complicating the form extraction process. This survey paper focuses on the methodologies employed for extracting form information from noisy scanned documents. Form understanding as well as more general document understanding encompass several distinct tasks, including document layout analysis, information extraction, and document visual question answering:

- **Document layout analysis (DLA)s** (Zhong et al. 2019): This task involves identifying and categorizing different regions of a document, such as text blocks, images, tables, and forms, to understand the document's structure.
- **Information extraction (IE)** (Zhang et al. 2020; Abdallah et al. 2024, 2023): This task focuses on extracting specific pieces of information from a document, such as names, dates, addresses, and other key-value pairs, often involving understanding the context and relationships between different pieces of text.
- **Document visual question answering (DocVQA)** (Mathew et al. 2021): This task involves answering questions related to the content of a document image by understanding both the visual and textual information present.In this paper, we present a comprehensive survey of research works on form understanding in the context of scanned documents. While our primary focus is on Transformer-based architectures, we also discuss other significant methodologies, such as graph-based approaches, to provide a holistic view of the field. Our research methodology involves an in-depth analysis of popular documents and trends over the last decade, enabling us to offer insights into the evolution of this domain. Focusing on cutting-edge models, we showcase how Transformers have propelled the field forward, revolutionizing form-understanding techniques. Our exploration includes an extensive examination of state-of-the-art language models designed to effectively tackle the complexities of noisy scanned documents. Furthermore, we present an overview of the latest and most relevant datasets, which serve as essential benchmarks for evaluating the performance of selected models. By comparing and contrasting the capabilities of these models, we aim to provide researchers and practitioners with useful guidance in choosing the most suitable solutions for their specific form understanding tasks.

The main contributions of this paper can be summarized as follows:

1. Comprehensive survey of recent advancements in form understanding for scanned documents, highlighting the role of language models and transformers as well as graph-based approaches.
2. In-depth analysis of cutting-edge transformer models, focusing on their effectiveness in handling noisy scanned documents. Comparative assessment and practitioner guidance were provided.

3. Overview of essential datasets serving as benchmarks for evaluating model performance. Identification of research gap and contribution through a dedicated form understanding survey.The structure of our paper is as follows: In Sect. 2, we present our research methodology, detailing how we collected and curated papers for this survey. We delve into different approaches to address form understanding in noisy scanned documents in Sect. 3. In Sect. 4, we describe commonly used datasets, which serve as the foundation for comparing various approaches. The comparative analysis is presented in Sect. 5.2. Finally, we offer an outlook on potential future directions and conclude in Sect. 6.

2 Literature collection methodology

To ensure the comprehensiveness of our survey, we employed a rigorous methodology for collecting relevant literature in the field of form understanding in noisy scanned documents. Our goal was to cover recent advancements and capture the latest trends in this rapidly evolving domain.

We conducted searches in prominent scientific databases, namely Scopus, on Aug 6th, 2023. The search query was designed to be comprehensive, combining relevant terms to target the topic effectively: "document understanding" AND "form" OR "information extraction" AND "Invoices"). By incorporating "document understanding" AND "form" in the search string, we ensured a strong connection between the topics of form understanding and document understanding.

The search results revealed substantial publication numbers across the databases. Springer Link returned the highest number of publications (162), with 50 of them published in 2021 alone. ACM yielded 75 relevant publications, with a noticeable increase in publications after 2018. IEEE returned 8 relevant publications.

In addition to the initial search results, we included the LayoutLMv2 approach (Xu et al. 2021) as it represented one of the state-of-the-art models during our survey's timeline.

To ensure the thoroughness of our literature collection, we employed the backward-snowballing method on the initially collected papers. This approach helped us identify additional relevant papers that may not have appeared in the initial search results. Through this process, we selected highly relevant publications that showcased the latest advances in form understanding and introduced datasets dedicated to form understanding or document understanding as a whole.

The chart shown in Fig. 2 depicts the number of relevant publications per year. The x-axis represents the years from 2014 to 2023, while the y-axis denotes the count of publications. The bars provide a clear visual representation of the distribution of relevant literature over the years, with a notable surge in publications from 2019 to 2022. As illustrated in Fig. 2, the majority of the relevant literature emerged within the last three years. This trend aligns with the rise in popularity of the topic, primarily driven by the introduction of pre-trained language models like BERT (Subramani et al. 2020).

Our literature collection methodology ensures the inclusion of recent breakthroughs and facilitates a comprehensive survey that captures the most current state of form understanding in noisy scanned documents. The selected papers will serve as the foundation for the

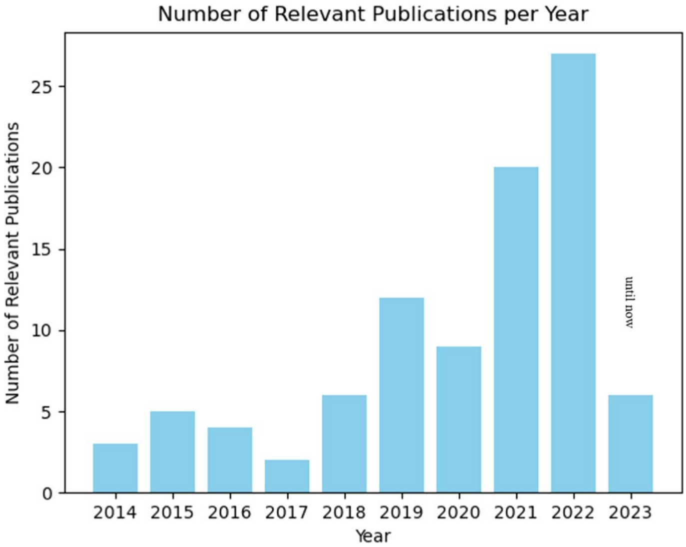


Fig. 2 Number of relevant publications per year

subsequent sections of this survey, where we delve into various approaches, datasets, and comparisons to shed light on the advancements in this vibrant research area.

3 Approaches

In recent years, the trajectory of models in the document understanding domain has been characterized by a paradigm shift towards enriching Transformer-based architectures with additional modalities beyond text. However, there are also significant approaches that do not rely on Transformers. This section introduces a series of significant models that have emerged in recent years, contributing to the advancement of document understanding. We present these models chronologically, illustrating their innovative integration of multi-modal information and their unique contributions to the field. A comprehensive overview of the covered models, including the modalities they leverage and the types of approaches they embody, is provided in Table 1.

3.1 Graph-based models

Graph-Based Text Relationship Modeling is a technique that employs graph neural networks (GNNs) to capture intricate relationships between text segments in documents. GNNs can process data that can be represented as graphs, such as nodes and edges.[1] These models use an encoder to process both textual and visual features, followed by a graph module that constructs a soft adjacency matrix to represent pairwise relationships between segments. This approach is particularly effective for capturing long-range dependencies and context in

[1] https://distill.pub/2021/gnn-intro/.

Table 1 An overview of the covered models in terms of their used modalities

Model	Text	1D Pos.	2D Pos.	Image	Approach type
BERT	✓	✓			
LayoutLM	✓	✓	✓	✓	Image-based fine-tuning
UniLMv2	✓	✓	✓		seq2seq
PICK	✓	✓	✓	✓	Graph-learning
TRIE	✓	✓	✓	✓	Multi-task (OCR + IE)
LayoutLMv2	✓	✓	✓	✓	Spatial-aware self-attention
StructuralLM	✓	✓	✓	✓	Text-segment grouping
SelfDoc	✓	✓	✓	✓	Text-segment grouping
DocFormer	✓	✓	✓	✓	Discrete multi-modal
StrucText	✓	✓	✓	✓	Text-segment grouping
BROS	✓	✓	✓		Using relative 2D positions
StrucTexTv2	✓	✓	✓	✓	Multi-modal fusion
Fast-StrucTexT	✓	✓	✓	✓	Efficient transformer
DocFormerv2	✓	✓	✓	✓	Multi-modal encoder
UDoc	✓	✓	✓	✓	Gated cross-attention
TILT	✓	✓	✓	✓	Text-image-lay-out transformer
GenDoc	✓	✓	✓	✓	Modality fu-sion, disentan-gled attention
MCLR	✓	✓	✓	✓	Cell-based lay-out, multi-scale layout
LiLT	✓	✓	✓	✓	Language-inde-pendent layout
FormNet	✓	✓		✓	Structural-aware attention
LayoutGCN	✓	✓	✓	✓	Lightweight GCN
GraphLayoutLM	✓	✓	✓	✓	Layout struc-ture modeling
UniLMv2	✓	✓	✓		seq2seq

documents with complex textual interactions, such as relation extraction and entity linking (Fu et al. 2019; Pham et al. 2023).

PICK (Yu et al. 2021) uses text and layout features in conjunction with a graph learning module to capture relationships between text segments. The architecture of PICK[2] is built

[2] https://github.com/wenwenyu/PICK-pytorch.

on three separate modules: The *encoder module* takes the original document as input. For each sentence of a document, the input vector consists of the text embeddings of characters, the bounding box, and the image embedding. Text segments are encoded using a transformer (Vaswani et al. 2017) and image segments are encoded using a pre-trained ResNet CNN (He et al. 2016). The text and image features are then combined by element-wise addition. The *graph module* takes the combined representation of the encoder module and layout bounding boxes of the original document as input. It is trained to generate a *soft adjacent matrix* that captures pairwise relationships between two graphs nodes. The usage of graph learning allows this model to capture long-distance relationships between text segments, which is traditionally a problem for transformer-based language models.

The *decoder module* that takes both the results of the encoder and the results of the graph module as input and forms the final output. The decoder uses a bidirectional long short-term memory network (BiLSTM) and a conditional random field algorithm (CRF) to perform sequence tagging on the union non-local sentence at a character level.

The *Text Reading and Information Extraction* (**TRIE**) model (Zhang et al. 2020) is unique in that the usually separated tasks of text-reading and text-understanding are fused into an end-to-end trained network. The general assumption is that both tasks can mutually benefit from each other during pre-training. The model consists of a *text-reading module*, a *multi-modal context block* and an *information-extraction module*. The *text-reading module* uses a ResNet CNN (Xie et al. 2017) in combination with a Feature Pyramid Network (FPN) (Lin et al. 2017) to extract the convolutional features I for an input image. This image embedding is passed to a *detector* function (usually anchor-based or segmentation-based approaches), which determines possible text regions $B = (b_1, b_2, ..., b_m)$ as bounding boxes. The regions *B* are fused with the image features *I* using RoIAlign (He et al. 2017) to get the text-region (or visual) features $C = (c_1, c_2, ..., c_m)$. Each c_i is fed into an encoder/decode model to produce the corresponding characters $y = (y_1, y_2, ...y_T)$ where *T* is the length of the label sequence. These textual features are represented in textual feature vector *Z* which is used together with visual feature vector *C* for information extraction. The *multi-modal context block* fuses textual, positional, and visual features created by the previous task. It makes use of a multi-head self-attention mechanism for the text embeddings, creating a text-context vector $\tilde{C}$. The *information extraction module* uses the text-context $\tilde{C}$ and the visual context *C* to separate entities and the text-embedding *Z* to extract their entity values. First, the final context vector $\overline{c_i}$ is created as a weighted sum of the context vectors $\tilde{C}$ and *C*. This fused context vector is concatenated with the text-embedding, and the final feature vector is passed to bidirectional LSTM.

GraphLayoutLM (Li et al. 2023) is a method to enhance visually-rich document understanding by explicitly modeling the layout structure. This approach leverages graph-based techniques to capture the spatial and hierarchical relationships between different elements in a document. By representing document elements as nodes and their spatial relationships as edges, the model can effectively capture the structural layout of documents. This method is particularly useful for tasks requiring a deep understanding of the document's visual and spatial structure.

LayoutGCN (Shi et al. 2023) proposes a lightweight graph convolutional network (GCN) architecture designed to efficiently handle visually rich documents. LayoutGCN focuses on capturing the spatial layout and visual features of documents while maintaining computational efficiency. The model employs a simplified GCN structure that reduces the complexity and computational load, making it suitable for real-time applications. Despite its lightweight design, LayoutGCN achieves competitive performance by effectively modeling the relationships between different layout components.

3.2 Transformer-based models

Transformer-based models have revolutionized the field of natural language processing and have been extensively adapted for document understanding tasks. These models leverage the self-attention mechanism, allowing to capture dependencies across different parts of the input data regardless of their distance. This characteristic makes Transformers particularly powerful for tasks involving complex structures and large contexts, such as document layout analysis, information extraction, and document visual question answering. In recent years, various Transformer-based architectures have been developed and tailored to better handle the unique challenges presented by multimodal document data. These models not only process textual information but also integrate visual and layout features, resulting in a more comprehensive understanding of documents. The following sections discuss some of the most significant transformers models.

3.2.1 Layout-visual fusion models

Layout-Visual Fusion Models integrate layout and visual information from document images with textual content to enhance document understanding. Document understanding is automatically reading, analyzing, and extracting information from documents, such as forms, tables, invoices, etc1,2. Layout-Visual Fusion Models combine features such as bounding box coordinates, image embeddings, and spatial relationships to capture the structural layout of documents, making them suitable for tasks that require precise spatial information, such as form understanding and table extraction (Li et al. 2021; Zheng et al. 2023; Oral and Eryiğit 2022).[3][4]**LayoutLM** (Xu et al. 2020) was the first transformer model that jointly trained text-level document information (using WordPiece (Wu et al. 2016) representation) in conjunction with the text's 2D position representation. Additionally, layout information was added during fine-tuning using image embeddings of token regions. It achieved new state-of-the-art results for several downstream tasks including form understanding. Like basic BERT models, it consists of a pre-training and a fine-tuning task. For pre-training and fine-tuning, the text embedding of a word is used together with its two-dimensional position embedding (i.e. the bounding box of the word on the document). The image embedding of the respective section on the document image is added during fine-tuning.

LayoutLM uses two pre-training tasks:

- **Masked visual-language modeling (MVLM)** is inspired by the BERT task *Masked language modeling (MLM)*, where random tokens of a sentence are masked using a special token and the model tries to predict the correct word for the gap. LayoutLM extends this idea by masking the word, but keeping its positional information for the prediction of the missing word.

[3] 1https://github.com/microsoft/unilm.

[4] 2https://learn.microsoft.com/en-us/microsoft-365/syntex/document-understanding-overview.

- **Multi-label Document Classification** is an optional pre-training task, where the authors use document-labels from the IIT-CDIP dataset (Lewis et al. 2006) (discussed in Sect. 5) to guide the model to the correct document domain. Since this information is often not present for larger datasets, the authors propose that this task will be obsolete for future iterations.The authors fine-tune the model using a form understanding, a receipt understanding, and a document image classification task.

LayoutLMv2 (Xu et al. 2021) is the evolution of the LayoutLM model. The key idea is to further integrate the relation between text, layout, and visual information. The authors achieve this by incorporating image information during pre-training as well as the introduction of new pre-training tasks.

The vanilla LayoutLM model relied on the traditional transformer self-attention mechanism, which has been replaced by the *spatial aware self-attention mechanism*. This mechanism is build to extend the original attention α_{ij} between the two tokens i and j by taking their 1D and 2D relations into account. With $\mathbf{b}^{(1D)}$ being the learnable bias in one-dimensional space and $\mathbf{b}^{(2D_x)}$, $\mathbf{b}^{(2D_y)}$ denoting the 2D biases for x and y direction respectively, the new attention value α'_{ij} is defined as

$$\alpha'_{ij} = \alpha_{ij} + \mathbf{b}^{(1D)}_{j-i} + \mathbf{b}^{(2D_x)}_{xj-xi} + \mathbf{b}^{(2D_y)}_{yj-yi}.$$

The output vector is then defined as the weighted average of the normalized spatial attention scores.For pre-training, LayoutLMv2 still uses the MVLM task of the original LayoutLM but adds two additional tasks. The first is *text image alignment* (TIA), where image regions of random tokens are covered and the model needs to predict the labels of covered and not-covered tokens.

The second addition is called *text image matching* (TIM). Here, the model needs to predict whether text-information belongs to a given image. Negative examples for this task are acquired by replacing a document image with another one from the training set or by dropping it.

LayoutLMv3 (Huang et al. 2022) represents the latest evolution in the LayoutLM series, building upon the foundations laid by its predecessors, LayoutLM and LayoutLMv2. This model continues to enhance the integration of text, layout, and visual information, further refining the capabilities for document understanding tasks. LayoutLMv3 significantly improves the multi-modal fusion process by introducing a more sophisticated mechanism to combine textual and visual features. It leverages an enhanced attention mechanism that better captures the interplay between text and image modalities, leading to improved performance on tasks that require understanding of both content and layout. The attention mechanism in transformers allows the model to weigh the importance of different tokens in the sequence, and LayoutLMv3 extends this by incorporating visual context into the attention weights, thus facilitating a more holistic understanding of the document.

The pre-training tasks in LayoutLMv3 have been expanded to include more challenging objectives that force the model to learn deeper associations between the modalities. In addition to MVLM and TIA, LayoutLMv3 introduces a Segment Prediction Task, which requires the model to predict the correct sequence of text segments within a document, helping it to better understand document structure. Another new task is Image-Text Matching with Noise, where the model learns to robustly match text to its corresponding image region by introducing noise into the document images, enhancing its ability to deal with noisy or imperfect document scans. These tasks are theoretically grounded in the concept of multi-task learning, where learning multiple related tasks simultaneously can lead to better generalization and more robust models. Building on the spatial-aware self-attention mechanism of LayoutLMv2, LayoutLMv3 introduces further refinements to this mechanism, allowing it to more effectively leverage spatial relationships between text tokens. The spatial-aware self-attention mechanism enhances traditional self-attention by adding positional biases that account for the two-dimensional (2D) layout of text in documents. This leads to improved accuracy in tasks that depend on the spatial arrangement of text, such as form understanding and table extraction. LayoutLMv3 incorporates more sophisticated visual features extracted from document images, utilizing advanced convolutional neural networks (CNNs) to provide richer and more detailed visual context. These features are then integrated with textual embeddings in a way that preserves the spatial integrity of the document. The integration of CNNs allows the model to capture local visual patterns and details, which are crucial for understanding complex document layouts.

3.2.2 Multi-modal fusion models

Multi-Modal Transformer Approaches are a type of large models that advance the fusion of modalities by introducing novel mechanisms to better integrate text, layout, and visual information. These models often propose modifications to the transformer architecture, such as spatial-aware self-attention mechanisms, to capture multi-dimensional relationships between tokens. They also introduce new pre-training tasks, like text-image matching and sequence length prediction, to enhance the understanding of document content and layout. These models are suitable for various multimodal tasks, such as document classification, image captioning, and video understanding (Xu et al. 2023; Feng et al. 2022; Wang et al. 2023).

SelfDoc (Li et al. 2021) is a novel model in the way the textual features are encoded. Instead of working on word-level embeddings, SelfDoc receives text grouped by semantically relevant components also referred to as *document object proposals* (text-block, title, list, table, figure) as model input. For the component-specific text extraction, the authors train a Faster R-CNN (Ren et al. 2015) on public document datasets PubLayNet (Zhong et al. 2019) and DocVQA (Mathew et al. 2021) on a document object detection task. The categorized regions are cropped from the document and passed to an OCR engine.

The encoded visual and textual feature vectors are then passed separately through two distinct multi-head attention encoders. Since those encoders do not share information between the visual and textual modality, they are referred to as *single-modality encoders*. The resulting attention matrix of both modalities is then combined using a third encoder, making the system a *cross-modality encoder*. This mechanism allows the SelfDoc model to use multi-modal document features during the pre-training phase.

UniLMv2 (Bao et al. 2020) allows for pre-training of a unified language model consisting of an auto-encoding and and a partially auto-regressive model. The auto-encoding model predicts tokens by conditioning on context just like BERT (Devlin et al. 2019). The partially auto-regressive model is based on masked- and pseudo-masked tokens. It can predict one or more masked (or pseudo-masked) tokens per factorization step. This approach allows the model to train on masked-token *spans* instead of single tokens. The authors refer to this as *Pseudo-Masked Language Modelling*. Masking of the data is done as follows: 15% of original tokens are masked. 40% of the time n-gram blocks are masked and 60% of the time single tokens are masked. The pre-training objective is to minimize the summed cross-entropy loss for the reconstruction of masked- and pseudo-masked-tokens in both models. The model's input consists of word token-embeddings, their absolute position embedding, and their segment embedding.

DocFormer (Appalaraju et al. 2021) improves on multi-modal training by fusing vision and language network layers. The visual features are obtained by using a ResNet CNN (Xie et al. 2017) at a lower-resolution embedding level. The visual embedding is then reduced in dimensionality by applying a 1 x 1 convolution. For the text embedding, DocFormer uses the OCR result from the document image and tokenizes it using WordPiece (Wu et al. 2016). The tokenized text t_{tok} is in the form of $[CLS], t_1, t_2, ..., t_n$ where N = 511 and all further tokens t_i with $i > 511$ are ignored. The text embedding is padded using a [PAD] token that is ignored by the self-attention mechanism. The resulting text input feature is of the same shape as the visual feature and formally defined as $T = W_t(t_{tok})$ where the weight W_t is initialized from LayoutLM pre-training weights.

For the spatial features the authors use the top-left and bottom-right coordinate of each word's bounding box in addition to the box's width *w*, height *h*, the euclidean distance from each corner of a box to each corner of the next box, and the centroid distance between boxes. The latter two can be defined as $A_{rel} = A^{k+1}_{num} - A^{k}_{num}; A \in (x, y); num \in (1, 2, 3, 4, c)$ where *num* defines the direction of the relation (*c* stands for the centroid coordinate). Similar to many comparable models DocFormer also encodes a 1D position encoding P^{abs} which defines the token sequence.

StrucText (Li et al. 2021) model uses a multi-modal combination of visual and textual document features. The textual embedding consists of a language token embedding where each text sentence is processed coherently to preserve semantic context. This is different from other models where word tokens of a sentence are processed individually. A sentence S is a sorted set of words, with the line delimiters [CLS] and [SEP] inserted at the beginning (t_0) and the at the end (t_{n+1}) of a sentence respectively.

The final textual embedding is achieved by combining the language token embedding with the layout embedding (i.e. bounding box of token).

For extractions of features of an image I, ResNet (Xie et al. 2017) is used which creates the visual embedding. Additionally, a sequence id embedding is used to capitalize on text segment order.

For pre-training, the authors used MVLM of LayoutLM in combination with two additional tasks. The first novel task is called *sequence length prediction* (SLP) where the length of a text sequence has to be predicted based on the visual features of the document. This task forces the model to learn from the image embedding. The second additional task is *paired box direction* (PBD). For two text segments i and j, the model needs to predict in which direction segment j is located with respect to i. Here the 360-degree field is divided into eight buckets of identical size. PBD is therefore a classification task that forces the model to learn spatial text segment relationships.

BROS (Hong et al. 2022) model does not incorporate the image of the text for pre-training, which is considered SOTA. The authors claim that it is more effective to focus on the combination of text and position-information, ignoring the layout modality. BROS introduces a custom text serializer that sorts the tokens in their respective reading order.

The model's features do not include absolute 2D positions of text, but a relative position representation based on the position of neighboring tokens as shown in Fig. 5. This has the advantage that the model can capture relations between similar key-value pairs better compared to absolute spatial information.

3.2.3 Cross-Modal Interaction Models

Cross-Modal Interaction Models are a type of artificial intelligence model that facilitates interactions between different modalities like text and layout in multimodal data. They employ attention mechanisms and fusion strategies to enable cross-modal information exchange and enhance the perception and understanding of the data. These models are suitable for various cross-modal tasks, such as document analysis, image-text retrieval, and video question answering.

UDoc (Gu et al. 2023) is a framework that employs a multi-layer gated cross-attention encoder (Lu et al. 2019; Gu et al. 2020), enabling effective cross-modal interactions within a single model. Unlike traditional methods, UDoc takes into account the complex and diverse nature of documents by incorporating both textual and visual features. This approach addresses the challenges of hierarchical document structures, multimodal content, and spatial layout. UDoc employs three pretraining tasks: Masked Sentence Modeling (MSM) (Ren et al. 2015), Visual Contrastive Learning (VCL), and Vision-Language Alignment (VLA) (Chen et al. 2020). MSM predicts masked sentence embeddings, VCL focuses on quantized visual features, and VLA enforces alignment between textual and visual elements. The document images are processed through feature extraction and quantization, providing a structured representation. The pretraining tasks, including MSM, VCL, and VLA, contribute to enhanced embeddings, capturing both local and global dependencies.

TILT (Powalski et al. 2021) addresses the complex challenge of natural language comprehension in documents that extend beyond plain text by introducing the Text-Image-Layout Transformer (TILT) neural network architecture. TILT is designed to simultaneously capture layout information, visual features, and textual semantics, making it capable of handling documents with rich spatial layouts. TILT model integrates layout representation through attention biases and contextualized visual information, leveraging pre-trained encoder-decoder Transformers (Hewlett et al. 2016; Raffel et al. 2020). In the proposed TILT architecture, the model's strength lies in its ability to combine contextualized image embeddings (Ethayarajh 2019) with semantic embeddings. By incorporating both layout

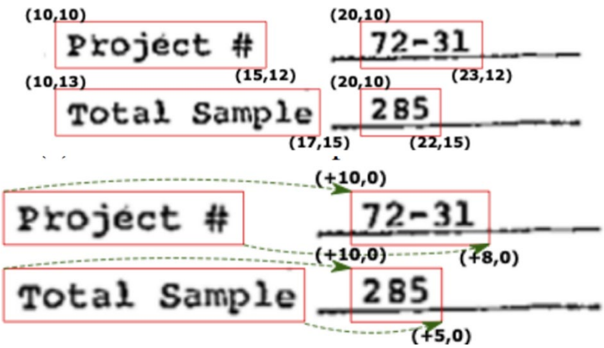


Fig. 5 Absolute (top) vs. relative (bottom) 2D position representation (Hong et al. 2022)

and visual information, TILT enhances its understanding of documents, particularly those with complex structures like forms and tables.

3.2.4 Sequence-to-sequence models

Encoder-Decoder and Sequence-to-Sequence Models involve encoders to process input data and decoders to generate sequential output. The encoders and decoders can be composed of different architectures, such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), or transformers. They are used for tasks like text generation, translation, and sequence prediction, where the input and output have different lengths or structures (Dalmia et al. 2019; Michael et al. 2019; Farina et al. 2023; Fu et al. 2023).

GenDoc (Feng et al. 2023) model employs an encoder-decoder architecture, allowing adaptability to diverse downstream tasks with varying output formats, which differentiates it from common encoder-only models. GenDoc's pre-training tasks encompass masked text prediction, masked image token prediction, and masked coordinate prediction, catering to multiple modalities. The model integrates modality-specific instructions, disentangled attention (Peng et al. 2022), and a mixture-of-modality-experts(MoE) (Bao et al. 2022) approach to effectively capture information from each modality.

The fusion of modalities within the encoder is achieved through disentangled attention, leading to improved layout effectiveness. The authors employ disentangled attentions for various interactions, including content-to-content, content-to-layout in the x-dimension, and content-to-layout in the y-dimension. This is represented by the following equations:

$$\begin{aligned}
A_{ij}^{cc} &= \boldsymbol{Q}i^{c}\boldsymbol{K}j^{c\intercal},\\
A_{ij}^{cx} &= \boldsymbol{Q}i^{c}\boldsymbol{K}\delta_x(i,j)^{x\intercal},\\
A_{ij}^{cy} &= \boldsymbol{Q}i^{c}\boldsymbol{K}\delta_y(i,j)^{y\intercal},\\
A_{ij} &= A_{ij}^{cc} + A_{ij}^{cx} + A_{ij}^{cy}.
\end{aligned}$$

Here, A_{ij} represents the resulting attention score between positions *i* and *j*; A_{ij}^{cc} is the conventional attention score from content query $\boldsymbol{Q}^c$ to content key $\boldsymbol{K}^c$ for positions *i* and *j*; A_{ij}^{cx} and A_{ij}^{cy} denote the disentangled attention scores from content to layout; $\delta_*(i,j)$ denotes the relative distance function between positions *i* and *j*.In contrast, modality separation within the decoder employs the MoE strategy, which activates task-specific experts for natural language, image token prediction, and coordinate prediction tasks. The pre-training tasks include text infilling, masked image token prediction, and masked coordinate prediction, employing a shared vocabulary.

DocFormerv2 (Appalaraju et al. 2023) comprises a multi-modal encoder (Chen et al. 2020) tailored to process both textual and spatial characteristics inherent in documents. The model adopts traditional transformer-based language modeling to handle textual content while introducing inventive pre-training tasks like token-to-line and token-to-grid predictions to capture document layout and structure as shown in Figs. 3 and 4. This approach enables the model to understand the spatial arrangement of document elements.

To effectively incorporate spatial information from document images, the authors introduce spatial features that encode critical elements such as position, size, and category. By

Fig. 3 Token-to-line (Appalaraju et al. 2023)

encoding these elements, DocFormerv2 gains an understanding of the document's structure and the relationships between different components.

A key strength of DocFormerv2 lies in its adept multi-modal fusion techniques (Li et al. 2022). By employing self-attention mechanisms, the model seamlessly integrates textual and spatial features, enabling it to comprehensively understand document content and layout. This fusion facilitates improved performance in a range of document understanding tasks.

FormNet (Lee et al. 2022) is designed to improve the extraction of information from complex form-like documents. These documents, which often contain tables, columns, and intricate layouts, pose challenges for traditional sequence models. FormNet addresses this issue by incorporating structural information into the sequence modeling process. FormNet contains two components: Rich Attention and Super-Tokens. Rich Attention enhances attention score calculation by considering both semantic relationships and spatial distances between tokens, enabling the model to capture the underlying layout patterns more effectively. Super-Tokens are constructed for each word in the form through graph convolutions, recovering local syntactic relationships that may have been lost during serialization. The information extraction problem is formulated as sequential tagging for tokenized words, with the goal of predicting key entity classes for each token. The authors build upon the Extended Transformer Construction (ETC) framework, which efficiently handles long sequences. ETC employs sparse global–local attention, and FormNet enhances positional encoding through Rich Attention.

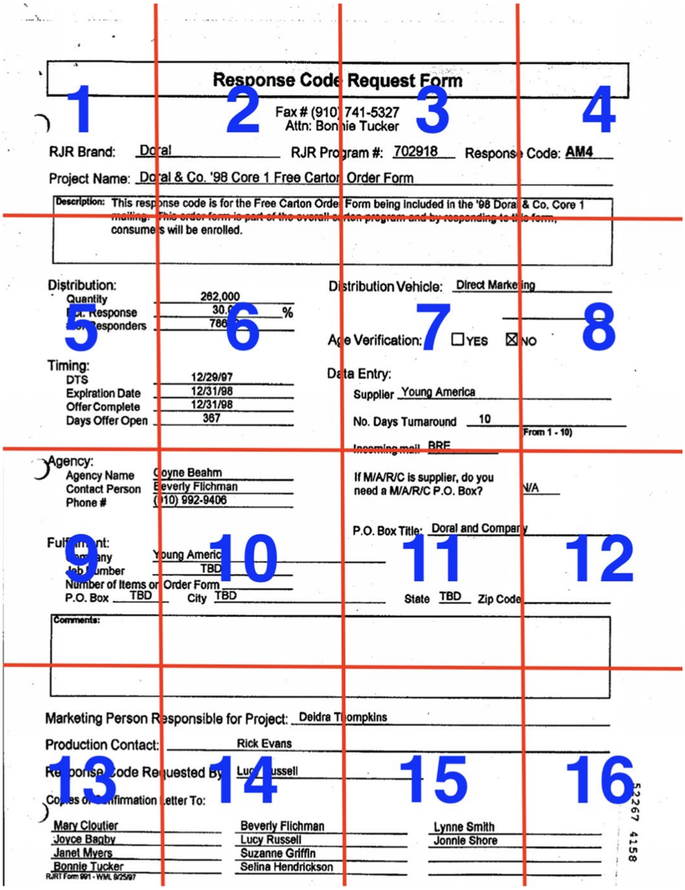


Fig. 4 Token-to-grid (Appalaraju et al. 2023)

3.2.5 Layout representation and language-independent models

Layout Representation Models (Hsiao et al. 2019; Tang et al. 2023; Wang et al. 2022) focus on capturing and representing the spatial arrangement of components in documents. They encode layout information for tasks like structure analysis and extraction. Language-Independent Models (Voutharoja et al. 2023; Gao et al. 2023; Nogueira et al. 2020) designed to work across multiple languages. They often utilize pre-training in one language and fine-tuning in others, enabling document understanding in various languages.

MCLR (Shi et al. 2023) proposed method employs a cell-based layout representation. This representation utilizes row and column indexes to define the position of components within a document, aligning with human reading habits and facilitating a more intuitive understanding of layout structures. The MCLR architecture comprises three main components: the cell-based layout, the multi-scale layout, and data augmentation.

In the cell-based layout, the traditional challenge of learning spatial relationships from coordinates is addressed by using row and column indexes. This enables the model to learn whether components share the same row or column and the number of components in each row or column. The approach also provides a more human-centered representation, aligning with natural reading patterns and enhancing the detection of latent rules within the document layout. Furthermore, the authors introduce a multi-scale layout representation by employing word- and token-level cells.

LiLT (Wang et al. 2022) introduces an innovative solution to a critical challenge in structured document understanding (SDU). The authors propose the Language-independent Layout Transformer (LiLT), a novel framework that enables pre-training on documents from a single language and subsequent fine-tuning for use with multiple languages.

The core architecture of LiLT is founded upon a parallel dual-stream Transformer model that efficiently processes both text and layout information. During pre-training, LiLT decouples the text and layout aspects of the document and employs a bi-directional attention complementation mechanism (BiACM) to facilitate interaction between these modalities. The model is then fine-tuned with a focus on downstream tasks using off-the-shelf pre-trained textual models. This process ensures that LiLT learns and generalizes the layout knowledge from monolingual documents to multilingual contexts.

3.2.6 Hybrid transformer architectures

Hybrid Transformer Architectures (Li et al. 2021; Yu et al. 2023) leverage innovative designs to address the challenges of computational complexity and diverse structural representations in document understanding. These models often incorporate novel attention mechanisms, such as Symmetry Cross-Attention (SCA) (Zhai et al. 2023), and utilize hourglass transformer architectures with merging and extension blocks to efficiently process multi-modal information. These designs enhance both performance and efficiency in analyzing complex documents.

StructuralLM (Li et al. 2021) is a successor to the LayoutLM model. It differs from LayoutLM by operating on 2D cell position embeddings instead of 2D word embeddings. The core idea is to better capture word relations by grouping them. Each document d is represented by a sequence of cells $d = \{c_1, ..., c_n\}$ and each cell is represented by a sequence of words $c_i = \{w_i^1, ..., c_i^m\}$, where each word w_i of cell c_i has the same 2D position embedding. In addition to this, the 2D embedding receives 1D position information per token (i.e. the sequence number of the token) to leverage word order within a cell, and the embedding of the word itself using WordPiece (Wu et al. 2016).

The first pre-training task for the StructuralLM model is the MVLM task introduced by LayoutLM (Xu et al. 2020). Since positional information is not dropped during MVLM and words of the same cell share common position information, StructuralLM can leverage the cell grouping during this task.

The second pre-training task is called *cell position classification* (CPC). Here, a set of scanned documents is split into N areas. For a random cell of the documents, the model needs to predict where the cell belongs using its center coordinate. This is in essence a multinomial classification task where each label $[1, N]$ belongs to a document section.

StrucTexTv2 (Yu et al. 2023) a model that combines visual and textual information to effectively analyze document images. The proposed approach employs a unique pre-training strategy with two self-supervised tasks: Mask Language Modeling (MLM) (Devlin et al. 2018) and Mask Image Modeling (MIM) (Bao et al. 2021; Chen et al. 2022), where text region-level masking is utilized to predict both visual and textual content. The core architecture of StrucTexTv2 comprises two main components: a visual extractor (CNN) and a semantic module (Transformer). The visual extractor processes the input document image using convolutional layers, extracting visual features from different stages of downsampling. The semantic module employs the Transformer architecture, which takes the flattened visual features and transforms them into a 1D sequence of patch token embeddings. A relative position embedding is added to the token embeddings, and the standard transformer processes these embeddings to generate enhanced semantic features. The resulting feature maps are reshaped to 2D context feature maps and up-sampled with a Feature Pyramid Network (FPN) strategy (Lin et al. 2017).

Fast-StrucTexT (Zhai et al. 2023) is an innovative and efficient transformer-based framework designed for document understanding tasks. The proposed model addresses the challenges posed by the computational complexity of traditional transformers while efficiently handling the diverse structural representations within documents. Fast-StrucTexT incorporates a novel hourglass transformer architecture, modality-guided dynamic token merging, and Symmetry Cross-Attention (SCA) (Wang et al. 2022) to achieve both high performance and efficiency.

The SCA module functions as a two-way cross-attention mechanism that facilitates interaction between different modes. In this module, one set of model features serves as the query, while another set of features from a different mode serves as both the key and value. This arrangement enables the calculation of cross-attention in both visual and textual modalities. The SCA is defined by the following equations:

1. Cross-Attention (CA) between feature sets f_n and f_m is given by:

$$\mathrm{CA}(f_n|f_m) = W_o \cdot \sigma\left(\frac{F_q(f_n)F_k^T(f_m)}{\sqrt{d}}\right) \cdot F_v(f_m),$$

where F_q, F_k, and F_v are linear projections for query, key, and value respectively. σ represents the softmax function, d is the hidden size, and W_o is the output weight.2. SCA between feature sets f_n and f_m is defined as:

$$\mathrm{SCA}(f_n, f_m) = \{\mathrm{CA}(f_n|f_m), \mathrm{CA}(f_m|f_n)\}.$$

Fast-StrucTexT leverages an hourglass transformer design, integrating Merging-Blocks and Extension-Blocks to efficiently process multi-modal information and handle complex document structures. The Merging-Blocks aggregate tokens to capture higher-level context, enhancing efficiency without compromising semantic richness. Complementarily, Extension-Blocks recover merged tokens, ensuring fine-grained details are preserved. The modality-guided dynamic token merging mechanism selectively merges tokens based on relevance, enabling diverse structural representations.

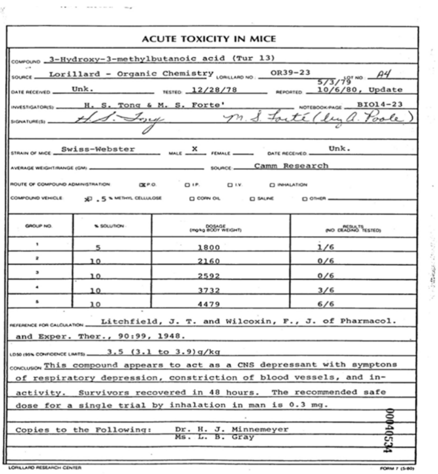
4 Datasets

In this section, we introduce commonly used datasets for form understanding tasks. For a comparison of basic features of the datasets, see Table 2.

Table 2 Comparison between individual datasets

Dataset	Document content	Color	Used languages	#Documents	Document type
FUNSD	Machine written/handwritten	B/W	EN	199	Form documents
XFUND	Machine written/handwritten	B/W	Multilingual	199/language	Form documents
NAF	Machine written pre-printed text and mostly handwritten input text	B/W	EN	865	Multiple
IIT-CDIP	Machine written, handwritten	B/W	EN	6, 919, 192	Multiple
RVL-CDIP	Machine written, handwritten	B/W	EN	400, 000	Multiple
PubLayNet	Machine written	Color	EN	364, 232	Medical literature
SROIE	Machine written with occasional handwriting	Color, B/W	EN	1000	Receipts
CORD	Machine written with occasional handwriting	Color	EN	1000 released, 11, 000 total	Receipts
DocVQA	Machine written, handwritten	Color, B/W	EN	12, 767	Multiple
Form-NLU	Machine written, handwritten	Color, B/W	EN	10, 857	Multiple
VRDU	Machine written	Color, B/W	EN	2, 556	Ad-buy

All listed datasets are publicly available



ACUTE TOXICITY IN MICE

COMPOUND 3-Hydroxy-3-methylbutanoic acid (Tur 13)
SOURCE Lorillard - Organic Chemistry LORILLARD NO. OR39-23 LOT NO. A4
DATE RECEIVED Unk. TESTED 12/28/78 REPORTED 5/3/79 10/6/80, Update
INVESTIGATOR(S) H. S. Tong & M. S. Forte' NOTEBOOK/PAGE BI014-23
SIGNATURE(S) H. S. Tong M. S. Forté (by A. Poole)

STRAIN OF MICE Swiss-Webster MALE X FEMALE DATE RECEIVED Unk.
AVERAGE WEIGHT/RANGE (GM) SOURCE Camm Research
ROUTE OF COMPOUND ADMINISTRATION X P.O. I.P. I.V. INHALATION
COMPOUND VEHICLE X .5 % METHYL CELLULOSE CORN OIL SALINE OTHER

GROUP NO.	% SOLUTION	DOSAGE (mg/kg BODY WEIGHT)	RESULTS (NO. DEAD/NO. TESTED)
1	5	1800	1/6
2	10	2160	0/6
3	10	2592	0/6
4	10	3732	3/6
5	10	4479	6/6

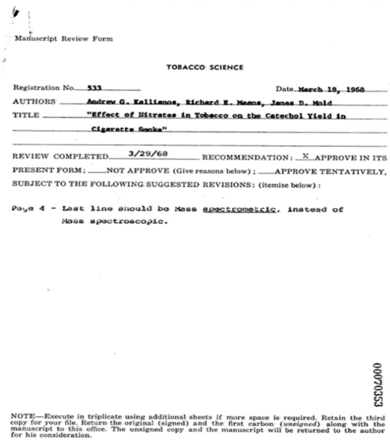
REFERENCE FOR CALCULATION Litchfield, J. T. and Wilcoxin, F., J. of Pharmacol. and Exper. Ther., 90:99, 1948.
LD50 (95% CONFIDENCE LIMITS) 3.5 (3.1 to 3.9)g/kg
CONCLUSION This compound appears to act as a CNS depressant with symptons of respiratory depression, constriction of blood vessels, and inactivity. Survivors recovered in 48 hours. The recommended safe dose for a single trial by inhalation in man is 0.3 mg.

Copies to the Following: Dr. H. J. Minnemeyer
Ms. L. B. Gray

00040534

LORILLARD RESEARCH CENTER FORM 7 (5-80)

(a)



Manuscript Review Form

TOBACCO SCIENCE

Registration No. 533 Date March 18, 1968
AUTHORS Andrew G. Kallianos, Richard E. Means, James D. Mold
TITLE "Effect of Nitrates in Tobacco on the Catechol Yield in Cigarette Smoke"

REVIEW COMPLETED 3/29/68 RECOMMENDATION: X APPROVE IN ITS PRESENT FORM; NOT APPROVE (Give reasons below); APPROVE TENTATIVELY, SUBJECT TO THE FOLLOWING SUGGESTED REVISIONS: (itemize below):

Page 4 - Last line should be Mass spectrometric, instead of Mass spectroscopic.

00070353

NOTE—Execute in triplicate using additional sheets if more space is required. Retain the third copy for your file. Return the original (signed) and the first carbon (*unsigned*) along with the manuscript to this office. The unsigned copy and the manuscript will be returned to the author for his consideration.

(b)

Fig. 6 Examples from the FUNSD dataset

RVL-CDIP and IIT-CDIP: The *Ryerson Vision Lab Complex Document Information Processing* RVL-CDIP dataset (Harley et al.) was created to evaluate Deep CNNs.[5] The dataset is a subset of the *IIT Complex Document Information Processing Test Collection* IIT-CDIP (Lewis et al. 2006, 2011) dataset, which itself contains a collection of documents from the Legacy Tobacco Documents Library (LTDL) created by the University of California San Francisco (Schmidt et al. 2002). IIT-CDIP consists of 6, 919, 192 document records in XML format, each of which contains text recognized by OCR and metadata information (document names, geographical information, etc.) of various structure and quality. Additionally, IIT-CDIP contains source images of scanned documents (Lewis et al. 2021).

In total, the RVL-CDIP dataset consists of 400, 000 grayscale images separated into 16 classes (including form documents) with 25, 000 images per class. The dataset is split into 320k training, 40k validation, and 40k test images Figs. (5, 6, 7, 8, 9, 10, 11, 12, 13).

FUNSD: The dataset for Form Understanding in Noisy Scanned Documents (FUNSD) (Jaume et al. 2019) is a collection of 199 fully annotated forms from various fields like marketing, science, and more.[6] All forms are rasterized, low-resolution one-page forms containing noise. More specifically, the authors manually selected 3, 200 eligible documents from the RVL-CDIP dataset (Lewis et al. 2006) form category and used random sampling to gain a final set of 199 documents. Most of the textual content is machine-written, which might not reflect real-life scenarios where handwritten content can be more prominent.

All annotations used for text detection were done by external contractors. The remaining tasks were annotated using an undisclosed annotation tool. Every form in the dataset is encoded in a JSON file where each form is considered as a list of interlinked *semantic entities* (i.e. a name field that is interlinked with its corresponding answer box). Each semantic

[5] https://www.cs.cmu.edu/~aharley/rvl-cdip/.

[6] https://guillaumejaume.github.io/FUNSD/.

Bezeichnung, Ort und Geschäftsnummer des Gerichts: ______

Erklärung über die persönlichen und wirtschaftlichen Verhältnisse bei Prozess- oder Verfahrenskostenhilfe
– Belege sind in Kopie durchnummeriert beizufügen –

A Angaben zu Ihrer Person

Julian Herrmann Name, Vorname, ggf. Geburtsname	Lehrer Beruf, Erwerbstätigkeit	09.08.1987 Geburtsdatum	ledig Familienstand
55232 Alzey, Bahnhofstraße 19 Anschrift (Straße, Hausnummer, Postleitzahl, Wohnort)		16724891042 Tagsüber tel. erreichbar unter Nummer	
Julia Herrmann, Julia, 55232 Alzey, Bahnhofstraße 30, 18757907013 Sofern vorhanden: Gesetzlicher Vertreter (Name, Vorname, Anschrift, Telefon)			

B Rechtsschutzversicherung/Mitgliedschaft

1. Trägt eine Rechtsschutzversicherung oder eine andere Stelle/Person (z. B. Gewerkschaft, Mieterverein, Sozialverband) die Kosten Ihrer Prozess- oder Verfahrensführung? Beleg Nummer

☐ Nein ☐ Ja: 123
In welcher Höhe? Wenn die Kosten in voller Höhe von einer Versicherung oder anderen Stelle/Person getragen werden, ist die Bewilligung von Prozess- oder Verfahrenskostenhilfe nicht möglich und damit die Beantwortung der weiteren Fragen nicht erforderlich. 34

2. Wenn nein: Besteht eine Rechtsschutzversicherung oder die Mitgliedschaft in einem Verein/einer Organisation (z. B. Gewerkschaft, Mieterverein, Sozialverband), der/die die Kosten der beabsichtigten Prozess- oder Verfahrensführung tragen oder einen Prozessbevollmächtigten stellen könnte? Beleg Nummer

☒ Nein ☐ Ja: Familienversicherung
Bezeichnung der Versicherung/des Vereins/der Organisation. Klären Sie möglichst vorab, ob die Kosten getragen werden. Bereits vorhandene Belege über eine (Teil-)Ablehnung seitens der Versicherung/des Vereins/der Organisation fügen Sie dem Antrag bei. 65

C Unterhaltsanspruch gegenüber anderen Personen

Haben Sie Angehörige, die Ihnen gegenüber gesetzlich zur Leistung von Unterhalt verpflichtet sind (auch wenn tatsächlich keine Leistungen erfolgen)? z. B. Mutter, Vater, Ehegatte/Ehegattin, eingetragene(r) Lebenspartner/Lebenspartnerin Beleg Nummer

☒ Nein ☐ Ja:
Name des Unterhaltsverpflichteten. Bitte geben Sie auf einem weiteren Exemplar dieses Formulars seine persönlichen und wirtschaftlichen Verhältnisse an, sofern diese nicht bereits vollständig aus den folgenden Abschnitten ersichtlich sind. 54

D Angehörige, denen Sie Bar- oder Naturalunterhalt gewähren

Name, Vorname, Anschrift (sofern sie von Ihrer Anschrift abweicht)	Geburts-datum	Verhältnis (z. B. Ehegatte, Kind, Mutter)	Monatsbetrag in EUR, soweit Sie den Unterhalt nur durch Zahlung gewähren	Haben diese Angehörigen eigene Einnahmen? z. B. Ausbildungsvergütung, Unterhaltszahlung vom anderen Elternteil usw.		Beleg Nummer
1 Max Weber	09.01.1991	Kind	0	☐ Nein	☒ Ja: 1323 mtl. EUR netto	23
2 Lilian Weber	03.04.1902	Mutter	1490	☐ Nein	☒ Ja: 231 mtl. EUR netto	34
3 Rudolf Schiff	05.06.1949	Vater	1589	☐ Nein	☐ Ja: 231 mtl. EUR netto	64
4				☐ Nein	☐ Ja: mtl. EUR netto	Cinesi sch
5				☐ Nein	☐ Ja: mtl. EUR netto	

- Allgemeine Fassung -
JV 205 (2) Erklärung über die persönlichen und wirtschaftlichen Verhältnisse bei Prozess- oder Verfahrenskostenhilfe (6.17)

(a)

Formular drucken

Antragsteller:
Max Weber

Empfänger:
Julian Weber

Antrag zum Kauf und Gebrauch von Kleinfeuerwerk der Klasse II

Ich beantrage / Wir beantragen die Freistellung vom Verwendungsverbot des § 23 (1) 1. Halbsatz gemäß § 24 (1) der 1. SprengV (Bekanntmachung 31.01.91, Bundesgesetzblatt BGBL I, S. 169).
Die Klassen III und IV sollen nicht mit abgebrannt werden, daher ist auch kein Pyrotechniker mit Erlaubnis gemäß § 27 oder Befähigungsschein nach § 20 SprengG erforderlich.
Ferner beantragen wir die zur Beschaffung des vorgesehenen Kleinfeuerwerks (Sonnen, Fontänen, Raketen etc.) notwendige Ausnahmegenehmigung gemäß § 24 (1) der 1. SprengV [siehe hierzu § 21 (1)]. Ich versichere / Wir versichern, daß das Abbrennen des Kleinfeuerwerks nicht in der Nähe von Anlagen und Gebäuden stattfindet, die in § 24 (1) der 1. SprengV als besonders schützenswert genannt sind.

Begründung (Anlass):
Hochzeit meiner Kinder

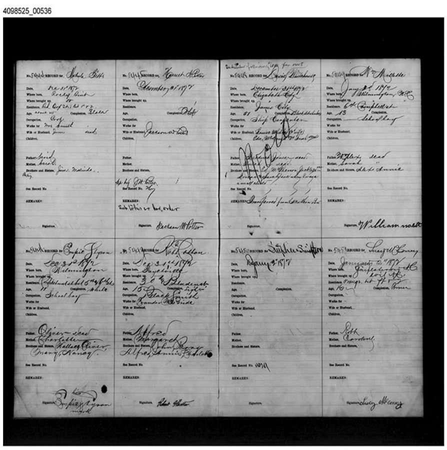
Veranstaltungsort / Tag / Uhrzeit (Zeitraum):
55232 Alzey, Bahnhofstraße 30
09.09.2019
um 11,00 Uhr

Durchführung durch (Name/Anschrift der verantwortlichen Person):
Max Weber, 55232 Alzey, Bahnhofstraße 30

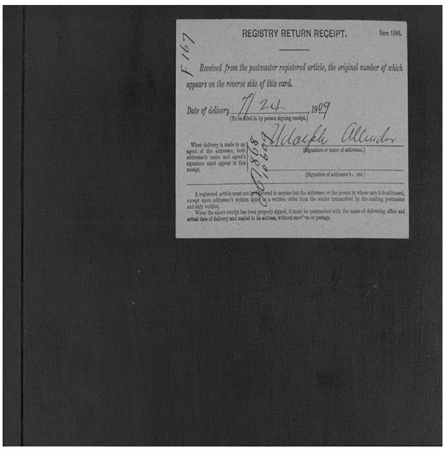
Ort, Datum: 07.08.1978 **Unterschrift:**

(b)

Fig. 7 Examples from the XFUNSD dataset



(a)



(b)

Fig. 8 Examples from the NAF dataset

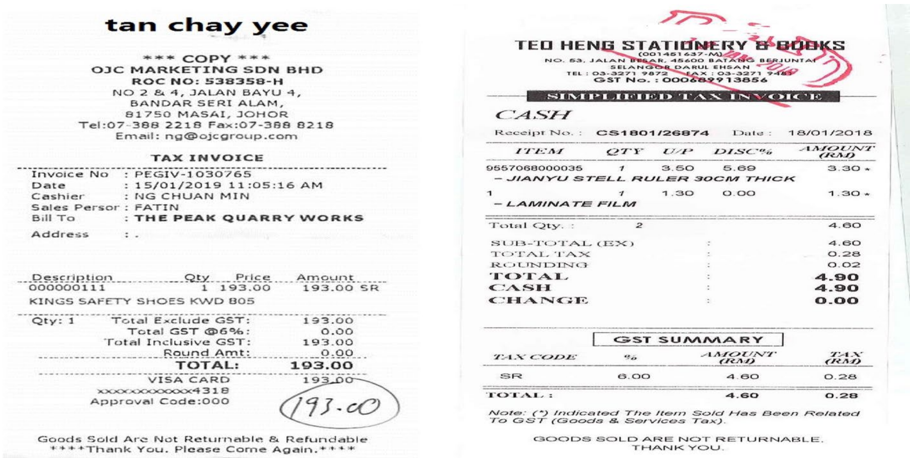
entity contains an identifier, a label, a bounding box, a list of links with other entities, and a list of words. In total, the dataset contains 9, 707 semantic entities and 5, 304 relations.

The authors of Vu and Nguyen (2020) have found several errors and inconsistencies in the FUNSD dataset including incorrect textual contents and bounding boxes as well as inconsistencies in annotating relations. They released an updated version of the dataset that fixes the discovered issues.[7]

[7] https://drive.google.com/drive/folders/1HjJyoKqAh-pvtg3eQAmrbfzPccQZ48rz?fbclid=IwAR2ouj5Sh0vkcKAMNfSoZjSM7vSpGnbK-AowWZZ8_Lltcn34hr7_nVcazu0.

(a) (b)

Fig. 9 Examples from the PubLayNet dataset


tan chay yee
*** COPY ***
OJC MARKETING SDN BHD
ROC NO: 538358-H
NO 2 & 4, JALAN BAYU 4,
BANDAR SERI ALAM,
81750 MASAI, JOHOR
Tel:07-388 2218 Fax:07-388 8218
Email: ng@ojcgroup.com

TAX INVOICE

Invoice No : PEGIV-1030765
Date : 15/01/2019 11:05:16 AM
Cashier : NG CHUAN MIN
Sales Person : FATIN
Bill To : THE PEAK QUARRY WORKS
Address : .

Description	Qty	Price	Amount
000000111	1	193.00	193.00 SR
KINGS SAFETY SHOES KWD 805			

Qty: 1 Total Exclude GST: 193.00
Total GST @6%: 0.00
Total Inclusive GST: 193.00
Round Amt: 0.00
TOTAL: 193.00
VISA CARD 193.00
xxxxxxxxxxxx4318
Approval Code:000
193.00

Goods Sold Are Not Returnable & Refundable
****Thank You. Please Come Again.****

TEO HENG STATIONERY & BOOKS
(001451637-M)
NO. 53, JALAN BESAR, 45600 BATANG BERJUNTAI
SELANGOR DARUL EHSAN
TEL : 03-3271 9872 FAX : 03-3271 7481
GST No. : 000683913856

SIMPLIFIED TAX INVOICE

CASH

Receipt No. : CS1801/26874 Date : 18/01/2018

ITEM	QTY	U/P	DISC%	AMOUNT (RM)
9557068000035	1	3.50	5.69	3.30 *
- JIANYU STELL RULER 30CM THICK				
1	1	1.30	0.00	1.30 *
- LAMINATE FILM				

Total Qty. : 2 4.60
SUB-TOTAL (EX) : 4.60
TOTAL TAX : 0.28
ROUNDING : 0.02
TOTAL : 4.90
CASH : 4.90
CHANGE : 0.00

GST SUMMARY

TAX CODE	%	AMOUNT (RM)	TAX (RM)
SR	6.00	4.60	0.28
TOTAL :		4.60	0.28

Note: (*) Indicated The Item Sold Has Been Related To GST (Goods & Services Tax).

GOODS SOLD ARE NOT RETURNABLE.
THANK YOU.

(a) (b)

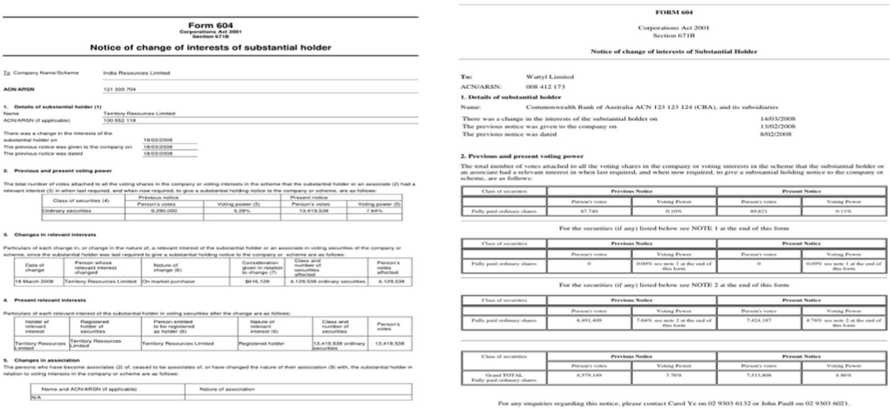
Fig. 10 Examples from the SROIE dataset

XFUND (Xu et al. 2021) is an extension to FUNSD where the dataset is translated into seven other languages, namely Chinese, Japanese, Spanish, French, Italian, German, and Portuguese. Like FUNSD, the dataset contains information on both semantic entities in a document and their relations to one another.[8] The dataset was manually created by gathering form templates in the above-mentioned languages from the internet. Each form is filled out exactly once with synthetic information to prevent repetition of forms and avoid shar-

[8] https://github.com/doc-analysis/XFUND.

(a) (b)

Fig. 11 Examples from the CORD dataset



Form 604
Corporations Act 2001
Section 671B
Notice of change of interests of substantial holder

To Company Name/Scheme India Resources Limited

ACN/ARSN 121 339 704

1. **Details of substantial holder (1)**
Name Territory Resources Limited
ACN/ARSN (if applicable) 100 552 118

There was a change in the interests of the substantial holder on 18/03/2008
The previous notice was given to the company on 18/03/2008
The previous notice was dated 18/03/2008

2. **Previous and present voting power**

The total number of votes attached to all the voting shares in the company or voting interests in the scheme that the substantial holder or an associate (2) had a relevant interest (3) in when last required, and when now required, to give a substantial holding notice to the company or scheme, are as follows:

Class of securities (4)	Previous notice		Present notice	
	Person's votes	Voting power (5)	Person's votes	Voting power (5)
Ordinary securities	9,290,000	5.29%	13,419,538	7.64%

3. **Changes in relevant interests**

Particulars of each change in, or change in the nature of, a relevant interest of the substantial holder or an associate in voting securities of the company or scheme, since the substantial holder was last required to give a substantial holding notice to the company or scheme are as follows:

Date of change	Person whose relevant interest changed	Nature of change (6)	Consideration given in relation to change (7)	Class and number of securities affected	Person's votes affected
18 March 2008	Territory Resources Limited	On market purchase	$616,128	4,129,538 ordinary securities	4,129,538

4. **Present relevant interests**

Particulars of each relevant interest of the substantial holder in voting securities after the change are as follows:

Holder of relevant interest	Registered holder of securities	Person entitled to be registered as holder (8)	Nature of relevant interest (6)	Class and number of securities	Person's votes
Territory Resources Limited	Territory Resources Limited	Territory Resources Limited	Registered holder	13,419,538 ordinary securities	13,419,538

5. **Changes in association**

The persons who have become associates (2) of, ceased to be associates of, or have changed the nature of their association (9) with, the substantial holder in relation to voting interests in the company or scheme are as follows:

Name and ACN/ARSN (if applicable)	Nature of association
N/A	

FORM 604

Corporations Act 2001
Section 671B

Notice of change of interests of Substantial Holder

To: Wattyl Limited
ACN/ARSN: 008 412 173

1. Details of substantial holder

Name: Commonwealth Bank of Australia ACN 123 123 124 (CBA), and its subsidiaries

There was a change in the interests of the substantial holder on 14/03/2008
The previous notice was given to the company on 13/02/2008
The previous notice was dated 8/02/2008

2. Previous and present voting power

The total number of votes attached to all the voting shares in the company or voting interests in the scheme that the substantial holder or an associate had a relevant interest in when last required, and when now required, to give a substantial holding notice to the company or scheme, are as follows:

Class of securities	Previous Notice		Present Notice	
	Person's votes	Voting Power	Person's votes	Voting Power
Fully paid ordinary shares	87,740	0.10%	89,621	0.11%

For the securities (if any) listed below see NOTE 1 at the end of this form

Class of securities	Previous Notice		Present Notice	
	Person's votes	Voting Power	Person's votes	Voting Power
Fully paid ordinary shares	0	0.00% see note 1 at the end of this form	0	0.00% see note 1 at the end of this form

For the securities (if any) listed below see NOTE 2 at the end of this form

Class of securities	Previous Notice		Present Notice	
	Person's votes	Voting Power	Person's votes	Voting Power
Fully paid ordinary shares	6,491,409	7.66% see note 2 at the end of this form	7,424,187	8.76% see note 2 at the end of this form

Class of securities	Previous Notice		Present Notice	
	Person's votes	Voting Power	Person's votes	Voting Power
Grand TOTAL Fully paid ordinary shares	6,579,149	7.76%	7,513,808	8.86%

For any enquiries regarding this notice, please contact Carol Ye on 02 9303 6132 or John Paull on 02 9303 6021.

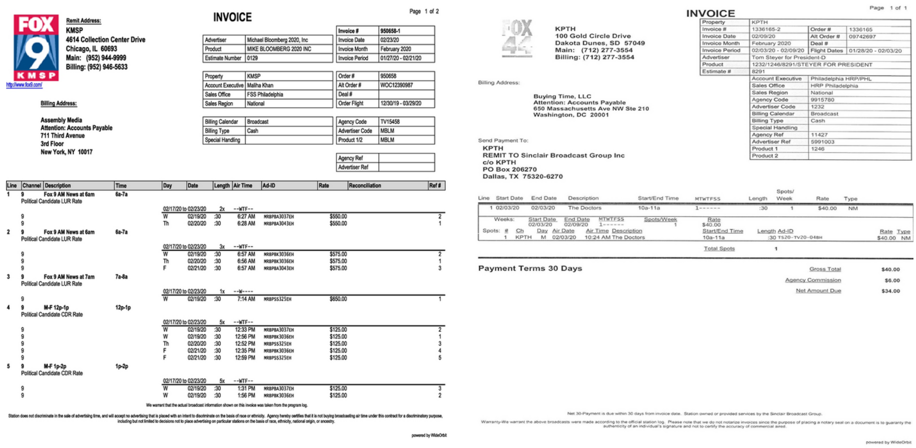
(a) (b)

Fig. 12 Examples from the Form-NLU dataset

ing sensitive information. Forms are filled out either digitally or with handwriting and then scanned to again be similar to the original FUNSD dataset.

To generate bounding boxes for each semantic entity, the authors use the Microsoft Read API (Microsoft 2022). Relation extraction is then done manually by annotators. In total, the XFUND dataset contains 199 forms per language (1, 393 total) split into training/test splits of 149/50 documents respectively.

NAF (Davis et al. 2019, 2019) is an annotated dataset created from historical form images of the United States National Archives. The forms have varied layouts and are noisy from degradation and machinery that was used to print them. A form in the dataset typically contains English pre-printed text and English input text consisting of handwriting,



(a) (b)

Fig. 13 Examples from the VRDU dataset

typed text, and stamped content. In total, the dataset contains 865 annotated grayscale form images. It is undisclosed how they were annotated.

The NAF dataset is manually split into a training, a validation, and a test set such that each set contains images with distinct form layouts. With this, the authors create a scenario where a classifier trained with the training set will not have seen the form layouts of the test set.

PubLayNet (Zhong et al. 2019) is a dataset that was *automatically* created by matching the XML representations and the content of 1, 162, 856 scientific PDF articles from PubMed Central^{TM}. The dataset was created primarily to advance research in *document layout analysis*, meaning it is not specifically made for form detection.[9] The annotations were created by parsing the PDF representation of the article using the PDFMiner (Shinyama 2019) software and then matching the generated layout with the respective XML representation.

While not mentioned in the paper, most of the annotated documents seem to have been removed during the quality control procedure as only a total of 364, 232 documents remain in the dataset after this step. The dataset is split into 340, 391 pages for training, 11, 858 pages for development, and 11, 983 pages for testing. Both development and testing sets have been selected from multiple different journals and contain a sufficient number of tables, lists, and figures.

The **SROIE** (Huang et al. 2019) dataset was created as part of the ICDAR 2019 competition on *Scanned Receipt OCR and Information Extraction*.[10] The dataset contains 1000 scanned receipt images with different annotations depending on one of three competition tasks (*Text Localisation*, *OCR*, and *Key-Information Extraction*). Receipts in the dataset may have poor paper, ink, or print quality, be of low resolution, and contain scanning artifacts. Further, receipts may contain unneeded interfering texts in complex layouts, long texts, and small font sizes. Each receipt contains English character text fields like goods

[9] https://github.com/ibm-aur-nlp/PubLayNet.

[10] https://drive.google.com/drive/folders/1ShItNWXyiY1tFDM5W02bceHuJjyeeJl2.

name and digit fields like unit price and total cost. The paper does not mention the origin of the receipts.

CORD: The *Consolidated Receipt Dataset* for post-OCR parsing (Park et al. 2019a) was created to be the first public dataset containing box-level text and parsing class annotations. Parsing class annotations are represented as a combination of one of eight superclasses (i.e. *store*, *payment*, and *menu*) and their respective subclasses (i.e. *name*, *address*, and *price*); e.g., *menu.price*.

The dataset itself contains 11, 000 Indonesian receipts from shops and restaurants gathered through crowdsourcing. Images were manually annotated using a web-based annotation tool where each image is first annotated and then checked for correctness and compliance with the annotation guidelines. Sensitive information like credit card numbers or a person's full name have been blurred in the final set of receipts. Out of the 11, 000 receipts, only 1, 000 have been made available to the public due to concerns of accidentally publishing sensitive personal data. According to one of the authors of the paper, future data release is unclear (Park et al. 2019b).

DocVQA: (Mathew et al. 2021, 2021) is a dataset of 12, 767 document images of varied types for the task of *Visual Question Answering*. The documents are sourced from documents of the UCSF Industry Documents Library (UCSF Library). Individual documents were hand-picked to provide suitable image quality and tables, forms, lists, and figures over long-running text fields. The documents are split randomly in an 80/10/10 ratio of train/validation/test splits respectively.

A total of 50, 000 questions and answers have been defined on the documents by remote workers using a three-step web-based annotation tool. In the first step, a worker defines up to ten question-answer pairs for a document. In the second step, another worker will try to answer the defined questions without seeing the pre-defined answers. If no answers of the first step match with any of the answers of the second step, the question is moved to a third step where the authors of the paper can manually edit the question-answer pairs.

Besides question-answer pairs, the dataset also contains one of nine question types per question. An example of a question type is *table/list* which specifies if table or list understanding is required to answer the question. A question can be assigned more than one question type.

Form-NLU: (Ding et al. 2023) is a comprehensive dataset designed to advance the field of Natural Language Understanding (NLU) in the context of form-based documents. The dataset comprises a total of 857 document images, 6k form keys and values, and 4k table keys and values. These documents encompass a variety of form types, including digital, printed, and handwritten. Form-NLU constitutes a portion of the publicly accessible financial form dataset originating from Form 604, which was amassed by SIRCA[11]. It provides text records of substantial shareholder notifications filed with the Australian Stock Exchange (ASX)[12] from 2003 to 2015. The authors segmented the annotation into three specialized sub-tasks: Key annotation for 12 types of Keys, Value annotation for paired Values, and Non-Key-Value annotation for other components like Title, Section, and Others. Each annotator handled one sub-task.

[11] https://asic.gov.au/regulatory-resources/forms/forms-folder/604-notice-of-changeof-interests-of-substantial-holder/.

[12] https://www2.asx.com.au/.

VRDU: is a benchmark dataset for Visually-rich Document Understanding, consisting of two distinct datasets: Ad-buy Forms, comprising 641 documents, and Registration Forms, comprising 1,915 documents. These datasets are designed to address challenges in extracting structured data from visually complex documents. The benchmark incorporates five key desiderata: rich schema, layout-rich documents, diverse templates, high-quality OCR results, and token-level annotation. The benchmark focuses on three tasks:

1. Single Template Learning (STL): the model is trained and evaluated on documents that belong to a single template. This means that the training, validation, and testing sets all contain documents that share the same structural layout or template. The goal of this task is to assess the model's ability to extract structured data when presented with a consistent and familiar document layout.
2. Mixed Template Learning (MTL): the model is trained and evaluated on documents that come from a set of templates. The training, validation, and testing sets include documents from different but predefined templates. This task evaluates the model's capacity to generalize its learning to diverse document layouts and templates that it has encountered during training.
3. Unseen Template Learning (UTL): The Unseen Template Learning task challenges the model to generalize beyond its training experience. In this task, the model is trained on documents from a subset of templates and is then evaluated on documents with templates it has never encountered during training. The goal is to assess the model's ability to adapt to new, previously unseen document layouts and templates.

The evaluation toolkit includes a type-aware fuzzy matching algorithm to assess extraction performance. Repeated, unrepeated, and hierarchical entity names are included, with post-processing heuristics to handle unrepeated and hierarchical entities.

5 Experiments results

In this section, we provide a comprehensive comparison of selected models on various datasets, including FUNSD, SROIE, CORD, RVL-CDIP, DocVQA, and Publaynet. We evaluate the models' performance using a range of metrics, including Precision, Recall, F1 Score, Accuracy, Average Normalized Line Similarity (Anls), and Mean Average Precision (MAP).

5.1 Evaluation metrics

We utilize several evaluation metrics to assess the effectiveness of the models in extracting structured information from scanned documents. The key evaluation metrics are as follows:

5.1.1 Precision, recall, F1 score, and accuracy

Precision, Recall, F1 Score, and Accuracy are standard metrics used to evaluate the performance of models in classification tasks. These metrics are calculated as follows:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{1}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{2}$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Samples}} \tag{4}$$

where:

- True Positives (TP) are the correctly predicted positive samples.
- False Positives (FP) are the incorrectly predicted positive samples.
- False Negatives (FN) are the incorrectly predicted negative samples.
- True Negatives (TN) are the correctly predicted negative samples.
- Total Samples is the total number of samples in the dataset.

5.1.2 Average normalized levenshtein similarity (anls)

Anls (Biten et al. 2019) captures the OCR mistakes applying a slight penalization in case of correct intended responses, but badly recognized. It also makes use of a threshold of value 0.5 that dictates whether the output of the metric will be the ANLS if its value is equal to or bigger than 0.5 or 0 otherwise. The key point of this threshold is to determine if the answer has been correctly selected but not properly recognized, or on the contrary, the output is a wrong text selected from the options and given as an answer.

More formally, the ANLS between the net output and the ground truth answers is given by the following equation. Where N is the total number of questions, M total number of GT answers per question, a_{ij} the ground truth answers where $i = \{0, ..., N\}$, and $j = \{0, ..., M\}$, and o_{qi} be the network's answer for the ith question q_i:

$$\text{ANLS} = \frac{1}{N} \sum_{i=0}^{N} \left(\max_j s(a_{ij}, o_{qi}) \right) \tag{5}$$

where: where $s(\cdot, \cdot)$ is defined as follows:

$$s(a_{ij}, o_{qi}) = \begin{cases} 1 - \text{NL}(a_{ij}, o_{qi}), & \text{if } \text{NL}(a_{ij}, o_{qi}) < \tau \\ 0, & \text{if } \text{NL}(a_{ij}, o_{qi}) \geq \tau \end{cases} \tag{6}$$

5.1.3 Mean average precision (MAP)

MAP is a metric commonly used to evaluate the quality of ranked lists in information retrieval tasks. It involves calculating the Average Precision (AP) for each query and then averaging these values across all queries. MAP is also used in object detection to evaluate the performance of a model. In object detection, the goal is to identify and localize objects in an image or video. The MAP metric takes into account both the accuracy of the detections and the precision of the detections. The formula for calculating MAP is as follows:

$$\mathrm{MAP} = \frac{1}{Q}\sum_{q=1}^{Q}\mathrm{AP}(q) \tag{7}$$

where:

- Q is the total number of queries.
- $\mathrm{AP}(q)$ is the Average Precision for query q, calculated as the average precision of relevant documents at each position in the ranked list:

$$\mathrm{AP}(q) = \frac{1}{N_q}\sum_{k=1}^{N_q}\mathrm{Precision}(k) \times \mathrm{Relevance}(k) \tag{8}$$

where:

- N_q is the total number of retrieved documents for query q.
- $\mathrm{Precision}(k)$ is the precision at position k in the ranked list.
- $\mathrm{Relevance}(k)$ is an indicator function that takes the value 1 if the document at position k is relevant to query q, and 0 otherwise.

5.2 Model comparison

In this section, We present a comparison of selected models on the FUNSD, SROIE, CORD, RVL-CDIP, DocVQA, and Publaynet datasets in Tables 3, 4, 5, 8, 6, and 7. The table provides the Precision, Recall, F1 Score, Accuracy, Anls, MAP, and the number of parameters for each model on each dataset.

5.2.1 FUNSD dataset

The FUNSD dataset, renowned for form understanding tasks, serves as a crucial benchmark for evaluating various models' performance. A comprehensive analysis of the comparison results presented in Table 3 yields valuable insights into the models' abilities. Notably, the transition from LayoutLM to LayoutLMv2 highlights the impact of introducing spatial-aware self-attention, leading to improved F1 scores. Noteworthy performance is observed from LayoutLMv3 *BASE*, which outperforms its predecessors with an impressive F1 score

Table 3 Performance comparison of information extraction models on the FUNSD dataset. The bold font indicate the best results.

Model	Precision	Recall	F1	#Params
$\text{LayoutLM}_{\text{Base}}$	0.7597	0.8155	0.7866	113 M
$\text{LayoutLM}_{\text{Large}}$	0.7596	0.8219	0.7895	343 M
$\text{LayoutLMv2}_{\text{Base}}$	0.8029	0.8539	0.8276	200 M
$\text{BROS}_{\text{Base}}$	0.8116	0.8502	0.8305	110 M
StrucText	0.8568	0.8097	0.8309	107 M
$\text{DocFormer}_{\text{Base}}$	0.8076	0.8609	0.8334	183 M
SelfDoc	–	–	0.8336	–
$\text{LayoutLMv2}_{\text{Large}}$	0.8324	0.8519	0.8420	426 M
$\text{BROS}_{\text{Large}}$	0.8281	0.8631	0.8452	340 M
$\text{DocFormer}_{\text{Large}}$	0.8229	0.8694	0.8455	536 M
FormNet-A2	0.8521	0.8418	0.8469	217 M
$\text{StructuralLM}_{\text{Large}}$	0.8352	0.8681	0.8514	355 M
UDoc	–	–	0.8720	274 M
$\text{DocFormerv2}_{\text{Base}}$	0.8915	0.876	0.8837	232 M
$\text{DocFormerv2}_{\text{Large}}$	0.8988	0.8792	0.8889	750 M
$\text{StrucTexTv2}_{\text{Small}}$	–	–	0.8923	107 M
$\text{Fast-StrucTexT}_{\text{Linear}}$	–	–	0.8950	–
$\text{LayoutLMv3}_{\text{Base}}$	–	–	0.9029	133 M
$\text{Fast-StrucTexT}_{\text{ResNet-18}}$	–	–	0.9035	–
$\text{StrucTexTv2}_{\text{Large}}$	–	–	0.9182	238 M
$\text{LayoutLMv3}_{\text{Large}}$	–	–	0.9208	368 M
$\text{MCLR}_{\text{Large}}$	–	–	0.9352	368 M
$\text{MCLR}_{\text{Base}}$	–	–	0.9376	133 M
DiT_{Base}	0.9470	0.9307	0.9388	87 M
$\text{DiT}_{\text{Large}}$	**0.9452**	**0.9336**	**0.9393**	304 M

Table 4 Performance comparison of information extraction models on the SROIE dataset. The bold font indicate the best results.

Model	Precision	Recall	F1	#Params
$\text{LayoutLM}_{\text{Base}}$	0.9438	0.9438	0.9438	113 M
$\text{LayoutLM}_{\text{Large}}$	0.9524	0.9524	0.9524	343 M
PICK	0.9679	0.9546	0.9612	–
TRIE	–	–	0.9618	–
$\text{LayoutLMv2}_{\text{Base}}$	0.9625	0.9625	0.9625	200 M
$\text{LayoutLMv2}_{\text{Large}}$	0.9661	0.9661	0.9661	426 M
StrucText	0.9584	0.9852	0.9688	107 M
$\text{Fast-StrucTexT}_{\text{Linear}}$	–	–	0.9712	–
$\text{Fast-StrucTexT}_{\text{ResNet-18}}$	–	–	0.9755	–
$\text{TILT}_{\text{Base}}$	–	–	0.9765	230 M
$\text{TILT}_{\text{Large}}$	–	–	**0.9810**	780 M

of 0.9029. StructuralLM *LARGE* stands out with an F1 score of 0.8514, indicating the success of its approach. StrucTexTv2 *LARGE*, despite its lower parameter count, achieves a competitive F1 score of 0.9182, demonstrating its efficiency in form understanding tasks. Additionally, DiT *BASE* and DiT *LARGE* showcase promising results with F1 scores of 0.9388 and 0.9393, respectively.

Table 5 Performance comparison of document layout analysis models on publaynet dataset. The bold font indicate the best results.

Model	MAP	#Params	
StrucTexTv2$_{224*224}$	0.8590	–	
StrucTexTv2$_{512*512}$	0.9210	–	
UDoc	Publaynet	0.9390	–
GenDoc$_{Base}$	0.9440	–	
StrucTexTv2$_{960*960}$	0.9490	–	
DiT$_{Base}$	0.9450	87 M	
DiT$_{Large}$	0.9490	304 M	
LayoutLMv3$_{Base}$	**0.9510**	133 M	

Table 8 Performance comparison of document layout analysis models on DocVQA dataset. The bold font indicate the best results.

Model	Anls	#Params
LayoutLMv2$_{\text{Large}}$	0.7808	200 M
LayoutLMv3$_{\text{Large}}$	0.7876	133 M
GenDoc$_{\text{Base}}$	0.7883	–
LayoutLMv3$_{\text{Large}}$	0.8337	368 M
TILT$_{\text{Base}}$	0.8392	230 M
GenDoc$_{\text{Large}}$	0.8449	–
StructuralLM$_{\text{Large}}$	0.8610	355 M
LayoutLMv2$_{\text{Large}}$	0.8672	426 M
TILT$_{\text{Large}}$	0.8705	780 M
DocFormerv2$_{\text{Base}}$	0.8720	232 M
DocFormerv2$_{\text{Large}}$	**0.8784**	750 M

Table 6 Performance comparison of document classification models on the RVL-CDIP dataset. The bold font indicate the best results.

Model	Accuracy	#Params	
MCLR$_{\text{base}}$	0.9070	133 M	
BEiT$_{\text{Base}}$	0.9109	87 M	
DiT$_{\text{Base}}$	0.9211	87 M	
DiT$_{\text{Large}}$	0.9269	304 M	
StrucTexTv2$_{\text{Small}}$	0.9340	28 M	
UDoc	0.9364	274 M	
GenDoc$_{\text{Base}}$	0.9380	–	
SelfDoc	RVL-CDIP	0.9381	–
LayoutLM$_{\text{Large}}$	0.9442	160 M	
LayoutLM$_{\text{Large}}$	0.9443	390 M	
GenDoc$_{\text{Large}}$	0.9450	–	
StrucTexTv2$_{\text{Large}}$	0.9462	238 M	
TILT$_{\text{Base}}$	0.9525	230 M	
LayoutLMv2$_{\text{Base}}$	0.9525	200 M	
LayoutLMv3$_{\text{Base}}$	0.9544	133 M	
DocFormer$_{\text{Large}}$	0.9550	536 M	
TILT$_{\text{Large}}$	0.9552	780 M	
LayoutLMv2$_{\text{Large}}$	0.9564	426 M	
LayoutLMv3$_{\text{Large}}$	0.9593	368 M	
StructuralLM$_{\text{Large}}$	0.9608	355 M	
DocFormer$_{\text{Base}}$	**0.9617**	183 M	

Table 7 Performance comparison of document layout analysis models on CORD dataset. The bold font indicate the best results.

Model	Precision	Recall	F1	#Params
$\text{LayoutLM}_{\text{Base}}$	0.9437	0.9508	0.9472	113 M
$\text{LayoutLM}_{\text{Large}}$	0.9432	0.9554	0.9493	343 M
$\text{LayoutLMv2}_{\text{Base}}$	0.9453	0.9539	0.9495	200 M
$\text{TILT}_{\text{Base}}$	–	–	0.9511	230 M
$\text{LayoutLMv2}_{\text{Large}}$	0.9565	0.9637	0.9601	426 M
$\text{DocFormer}_{\text{Base}}$	0.9652	0.9614	0.9633	183 M
$\text{TILT}_{\text{Large}}$	–	–	0.9633	780 M
$\text{LayoutLMv3}_{\text{Base}}$	–	–	0.9656	133 M
$\text{GenDoc}_{\text{Base}}$	–	–	0.9659	–
$\text{Fast-StrucTexT}_{\text{Linear}}$	–	–	0.9665	–
$\text{DocFormerv2}_{\text{Base}}$	0.9751	0.9610	0.9680	232 M
$\text{GenDoc}_{\text{Large}}$	–	–	0.9697	–
$\text{DocFormer}_{\text{Large}}$	0.9725	0.9674	0.9699	536 M
$\text{Fast-StrucTexT}_{\text{ResNet-18}}$	–	–	0.9715	–
$\text{MCLR}_{\text{Base}}$	–	–	0.9723	133 M
FormNet-A3	0.9802	0.9655	0.9728	345 M
$\text{LayoutLMv3}_{\text{Large}}$	–	–	0.9746	368 M
$\text{MCLR}_{\text{Base}}$	–	–	0.9749	368 M
$\text{DocFormerv2}_{\text{Large}}$	0.9771	0.9770	0.9770	750 M
$\text{UDoc}_{\text{Large}}$	–	–	**0.9813**	274 M

5.2.2 SROIE dataset

The evaluation of various models on the SROIE dataset, which is centered around receipt understanding and key-information extraction tasks, provides insights into the capabilities of these models for form-based understanding. Table 4 presents a comprehensive comparison of the performance achieved by different models on this dataset.

LayoutLMv2 *BASE* and LayoutLMv2 *LARGE* demonstrate their prowess in the receipt understanding task. Both models achieve high precision, recall, and F1 scores of 0.9625 and 0.9661 respectively. StrucText exhibits strong recall with a score of 0.9852, showcasing its effectiveness in capturing key information from receipts. Its F1 score of 0.9688 indicates a balanced trade-off between precision and recall, making it a viable option for the SROIE dataset.

5.2.3 Publaynet dataset

The PubLayNet dataset, despite its primary focus on advancing document layout analysis, has garnered attention from researchers in the field of form understanding due to its extensive annotations and diverse scientific content. As showcased in Table 5, various models have been evaluated on the PubLayNet dataset. Among the evaluated models, LayoutLMv3 *Base* stands out with a high Mean Average Precision (MAP) of 0.9510. This suggests that the model's spatial-aware self-attention mechanism and architecture contribute to effectively capturing layout information, even in the context of intricate scientific documents. Similarly, DiT *LARGE* showcases promising results with a MAP of 0.9490, demonstrating its capability in processing diverse document layouts for form understanding. GenDoc *Base*, while not specifically designed for layout analysis, still achieves a competitive MAP of 0.9440, indicating its generalizability to document understanding tasks. UDoc also demonstrates its effectiveness in this domain, achieving a MAP of 0.9390. StrucTexTv2 models, operating at different resolutions (224×224, 512×512, and 960×960), exhibit varying levels of performance, with the 960x960 variant achieving the highest MAP of 0.9490.

5.2.4 RVL-CDIP dataset

The RVL-CDIP dataset offers a distinct dimension in the realm of form understanding, featuring multi-class single-label classification tasks. The comprehensive model comparison presented in Table 6 provides valuable insights into various models' performances. Among these models, the DocFormer architecture emerges as a standout performer, particularly DocFormer *Base*, which attains an exceptional accuracy score of 0.9617, positioning itself at the forefront. This accomplishment underscores the efficacy of the DocFormer approach in tackling the challenges posed by multi-class classification tasks within the form understanding domain.

A noteworthy observation arises from the juxtaposition of DocFormer *LARGE* and DocFormer *BASE*, accentuating the interplay between model complexity and performance enhancement. Although DocFormer *LARGE* achieves marginally lower accuracy than its *BASE* counterpart, it does so with a notable increase in the number of model parameters. This trade-off underlines the nuanced relationship between model size and the achieved accuracy, showcasing the balance that must be struck when considering practical deployment and computational efficiency.

Moreover, this dataset comparison affirms the competitive capabilities of other models, such as StructuralLM *LARGE*, LayoutLMv3, LayoutLMv2, and StrucTexTv2 *LARGE*, each contributing to the diverse landscape of form understanding. Furthermore, the inclusion of various baseline models like BEiT and DiT provides a comprehensive perspective on the state-of-the-art advancements in multi-class classification on the RVL-CDIP dataset.

5.2.5 CORD dataset

While the CORD dataset is more oriented towards receipt understanding, The models demonstrate a range of strengths in capturing different aspects of the documents, including textual content, spatial arrangement, and overall structure. A comprehensive analysis of the comparison results presented in Table 7. Notably, the LayoutLM variants, LayoutLM *BASE* and LayoutLM *LARGE* exhibit strong performance with F1 scores of 0.9472 and 0.9493, respectively, benefiting from their layout-aware self-attention mechanisms. DocFormer models, particularly DocFormer *BASE* and DocFormer *LARGE*, excel in capturing complex structural patterns and linguistic context, achieving impressive F1 scores of 0.9633 and 0.9699. The LayoutLMv2 models also perform well, with LayoutLMv2 *LARGE* achieving an F1 score of 0.9601. The Fast-StrucTexT models, including Fast-StrucTexT *Linear* and Fast-StrucTexT *ResNet-18*, demonstrate solid performance with F1 scores of 0.9665 and 0.9715, respectively, indicating the efficiency of their hourglass transformer architecture. UDoc *LARGE* emerges as a top performer, achieving an impressive F1 score of 0.9813, showcasing the effectiveness of its gated cross-attention mechanism in capturing both textual and spatial information. Models like TILT *Base* and GenDoc *Base* also show competitive performance, highlighting their potential in addressing document understanding challenges. MCLR *base* and MCLR *base* achieve strong F1 scores of 0.9723 and 0.9749, respectively, emphasizing the success of their cell-based layout representation. FormNet achieves a high accuracy of 0.9728, showcasing its proficiency in extracting information from complex form-like documents.

5.2.6 DocVQA dataset

The evaluation of various models on the DocVQA dataset provides insights into their effectiveness in addressing the challenges posed by document-based visual question answering. Table 8 presents a comprehensive comparison of the performance achieved by different models on this dataset. StructuralLM *LARGE* stands out as a strong performer, boasting an analysis score of 0.8610. This achievement underscores the model's ability to integrate structural information from documents and effectively answer questions based on visual and textual cues. Both LayoutLMv3 *Large* and LayoutLMv2 *Large* demonstrate their prowess in document understanding, achieving scores of 0.7876 and 0.8672 respectively. These models leverage advanced self-attention mechanisms to capture contextual and spatial relationships in documents, enabling them to tackle complex questions that involve both textual and visual components.

5.2.7 Overall insights

In this section, we provide a comprehensive analysis of various datasets and model comparisons, offering insights into the adaptability and effectiveness of models like LayoutLMv2 and DocFormer in form understanding tasks. Through this exploration, we have identified key trends and challenges that extend beyond individual datasets, illuminating the broader landscape of form understanding. One notable observation is the diverse range of strategies employed to tackle form-understanding tasks, as revealed by our comparisons of models across different datasets. For example, LayoutLMv2 consistently demonstrates strong performance, particularly in handling complex document layouts, thanks to its spatial-aware self-attention mechanism. This ability to capture and utilize spatial relationships in the document structure positions LayoutLMv2 as a highly effective model for a wide array of form-related tasks.

Similarly, DocFormer has shown remarkable adaptability across multiple datasets, consistently delivering competitive results. Its versatility and consistent performance across diverse tasks underscore its potential as a robust model for handling various document processing challenges. This adaptability is especially evident when dealing with unstructured and noisy scanned documents, where models must effectively generalize to different formats and layouts. In our analysis, we also emphasize the trade-offs between model complexity and performance. While larger models, such as LayoutLMv2 and DocFormer, tend to yield improved results, these gains come at the cost of significantly higher computational resources. For example, models with a larger number of parameters often offer marginal performance improvements that may not justify their increased resource demands. Therefore, we highlight the importance of evaluating the trade-offs between model size, computational efficiency, and performance when selecting models for specific tasks or deployment environments.

Dataset-specific model selection is another critical consideration. Certain datasets, such as SROIE, which focuses on receipt recognition, may benefit from models like TILT *Large* that are specifically designed for context-dependent scenarios. This underscores the importance of tailoring models to the unique characteristics of the dataset, as domain-specific models often excel in handling specific challenges related to form understanding. Real-world datasets, particularly those containing noisy or complex documents, introduce additional challenges such as layout variations, noise, and diverse document formats. These hurdles highlight the necessity for models that can robustly handle such complexities. Our analysis underscores the importance of developing models that are not only accurate but also resilient and adaptable to real-world challenges.

Finally, we observe the interrelation between document layout analysis, information extraction, and document visual question answering (DocVQA). Advances in layout analysis, for example, can significantly enhance the precision of information extraction techniques by providing more accurate contextual cues. Robust information extraction capabilities are similarly crucial for improving the performance of DocVQA systems, which rely on precise data to generate correct answers. This synergy between tasks suggests that integrated approaches, which combine the strengths of layout analysis, information extraction, and visual question answering, may offer the most promising advancements in document processing technologies.

6 Conclusions

Form understanding, especially in the context of scanned documents, presents numerous challenges that stem from the inherent complexities of document layouts, noisy inputs, and unstructured formats. Through this survey, we have highlighted recent advancements in the field, particularly the effectiveness of models like LayoutLMv2 and DocFormer, which incorporate spatial-aware self-attention mechanisms to better capture the relationships between text, layout, and visual features. However, several challenges remain that need further exploration. One of the primary challenges is the efficient integration of text-position encoding with layout information. While models like LayoutLMv2 and DocFormer have made significant progress in leveraging layout information through position embeddings, there is still a need for more advanced techniques that can dynamically adjust to the variability in document structures. Future models must be able to seamlessly integrate the positional and visual context with textual features to improve the extraction of relevant information, particularly in documents where layout plays a critical role.

Another challenge is the computational complexity associated with larger models, which often require substantial resources for both training and inference. This can hinder their deployment in resource-constrained environments, such as mobile applications or edge devices. Future research could focus on developing more efficient architectures that maintain high accuracy while reducing computational overhead, potentially by leveraging techniques such as model pruning, quantization, or knowledge distillation. In terms of future directions, integrating multi-modal learning to combine textual, visual, and positional cues presents a promising avenue. Additionally, exploring new methods for handling noisy and complex layouts in real-world documents, such as scanned forms with irregular structures, will be crucial for advancing the field. Innovations in self-supervised learning, which can reduce the need for extensive labeled datasets, may also play a pivotal role in future breakthroughs.

Lastly, improving the handling of out-of-distribution documents and developing models that generalize better across diverse datasets are key areas for future work. As the field progresses, we anticipate the development of hybrid approaches that combine traditional rule-based systems with machine learning models to enhance robustness and adaptability in handling various forms and documents. Future work in form understanding could involve tackling challenges like handling mixed layouts, adapting models to different languages, and addressing complex structures within forms. Developing techniques that seamlessly extract information from diverse form components while accommodating unstructured text could lead to more versatile models. Additionally, exploring domain adaptation strategies and incorporating user feedback for model refinement could enhance real-world applicability. The establishment of standardized benchmark datasets and evaluation metrics tailored to specific form understanding tasks would provide a solid foundation for advancing the field.

Funding Open access funding provided by University of Innsbruck and Medical University of Innsbruck.

References

Abdallah A, Abdalla M, Elkasaby M, Elbendary Y, Jatowt A (2023) Amurd: annotated multilingual receipts dataset for cross-lingual key information extraction and classification, arXiv preprint arXiv:2309.09800

Abdallah A, Abdalla M, Kasem M. S, Mahmoud M, Abdelhalim I, Elkasaby M, ElBendary Y, Jatowt A (2024) Coru: comprehensive post-ocr parsing and receipt understanding dataset, arXiv preprint arXiv:2406.04493

Appalaraju S, Jasani B, Kota BU, Xie Y, Manmatha R (2021) Docformer: end-to-end transformer for document understanding. In: 2021 IEEE/CVF international conference on computer vision, ICCV 2021, Montreal, October 10–17, 2021, IEEE, pp 973–983. https://doi.org/10.1109/ICCV48922.2021.00103

Appalaraju S, Tang P, Dong Q, Sankaran N, Zhou Y, Manmatha R (2023) Docformerv2: local features for document understanding, arXiv preprint arXiv:2306.01733

Appalaraju S, Tang P, Dong Q, Sankaran N, Zhou Y, Manmatha R (2024) Docformerv2: local features for document understanding. In: Proceedings of the AAAI conference on artificial intelligence, vol 38, pp 709–718

Bao H, Wang W, Dong L, Liu Q, Mohammed OK, Aggarwal K, Som S, Piao S, Wei F (2022) Vlmo: unified vision-language pre-training with mixture-of-modality-experts. Adv Neural Inform Process Syst 35:32897–32912

Bao H, Dong L, Piao S, Wei F (2021) Beit: Bert pre-training of image transformers, arXiv preprint arXiv:2106.08254

Bao H, Dong L, Wei F, Wang W, Yang N, Liu X, Wang Y, Gao J, Piao S, Zhou M et al (2020) Unilmv2: pseudo-masked language models for unified language model pre-training. In: International conference on machine learning, PMLR, pp 642–652

Binmakhashen GM, Mahmoud SA (2019) Document layout analysis: a comprehensive survey. ACM Comput Surv (CSUR) 52(6):1–36

Biten AF, Tito R, Mafla A, Gomez L, Rusiñol M, Valveny E, Jawahar CV, Karatzas D (2019) Scene text visual question answering. arXiv:1905.13648

Chen H, Jiang D, Sahli H (2020) Transformer encoder with multi-modal multi-head attention for continuous affect recognition. IEEE Trans Multimed 23:4171–4183

Chen X, Ding M, Wang X, Xin Y, Mo S, Wang Y,Han S, Luo P, Zeng G, Wang J (2022) Context autoencoder for self-supervised representation learning, arXiv preprint arXiv:2202.03026

Chen T, Kornblith S, Norouzi M, Hinton G (2020) A simple framework for contrastive learning of visual representations. In: International conference on machine learning, PMLR, pp 1597–1607

Dalmia S, Mohamed A, Lewis M, Metze F, Zettlemoyer L (2019) Enforcing encoder-decoder modularity in sequence-to-sequence models. arXiv:1911.03782

Davis B, Morse B, Cohen S, Price B, Tensmeyer C (2019) Deep visual template-free form parsing. In: 2019 International conference on document analysis and recognition (ICDAR), IEEE, pp 134–141

Davis B, Morse B, Cohen S, Price B, Tensmeyer C (2019) National archives forms dataset, https://github.com/herobd/NAF_dataset

Devlin J, Chang M.-W, Lee K, Toutanova K (2018) Bert: pre-training of deep bidirectional transformers for language understanding, arXiv preprint arXiv:1810.04805

Devlin J, Chang M, Lee K, Toutanova K (2019) BERT: pre-training of deep bidirectional transformers for language understanding. In: J Burstein, C Doran, T Solorio (eds.), Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, NAACL-HLT 2019, Minneapolis, June 2–7, volume 1 (long and short papers), association for computational linguistics, 2019, pp 4171–4186. https://doi.org/10.18653/v1/n19-1423

Ding Y, Long S, Huang J, Ren K, Luo X, Chung H, Han SC (2023) Form-nlu: dataset for the form language understanding, arXiv preprint arXiv:2304.01577

Ethayarajh K (2019) How contextual are contextualized word representations? Comparing the geometry of bert, elmo, and GPT-2 embedding. In: K Inui, J Jiang, V Ng, X Wan (eds.), Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing, EMNLP-IJCNLP 2019, Hong Kong, November 3–7, 2019, association for computational linguistics, pp 55–65. https://doi.org/10.18653/v1/D19-1006

Farina M, Pappadopulo D, Gupta A, Huang L, İrsoy O, Solorio T (2023) Distillation of encoder-decoder transformers for sequence labelling. arXiv:2302.05454

Feng C-M, Yan Y, Chen G, Xu Y, Shao L, Fu H (2022) Multi-modal transformer for accelerated mr imaging. arXiv:2106.14248

Feng S, Zhan T, Jie Z, Luong T. Q, Jin X (2023) Sequence-to-sequence pre-training with unified modality masking for visual document understanding, arXiv preprint arXiv:2305.10448

Fu T-J, Li P-H, Ma W-Y (2019) GraphRel: Modeling text as relational graphs for joint entity and relation extraction. In: Proceedings of the 57th annual meeting of the association for computational linguistics, association for computational linguistics, Florence pp 1409–1418. https://doi.org/10.18653/v1/P19-1136. https://aclanthology.org/P19-1136

Fu Z, Lam W, Yu Q, So A. M.-C, Hu S, Liu Z, Collier N (2023) Decoder-only or encoder-decoder? Interpreting language model as a regularized encoder-decoder. arXiv:2304.04052

Gao T, Yen H, Yu J, Chen D (2023) Enabling large language models to generate text with citations. arXiv:2305.14627

Gu J, Kuen J, Joty S, Cai J, Morariu V, Zhao H, Sun T (2020) Self-supervised relationship probing. Adv Neural Inform Process Syst 33:1841–1853

Gu J, Nenkova AN, Barmpalios N, Morariu VI, Sun T, Jain RB, Kuen JWY, Zhao H (2023) Unified pretraining framework for document understanding, uS Patent App. 17/528,061 (May 18)

Han K, Wang Y, Chen H, Chen X, Guo J, Liu Z, Tang Y, Xiao A, Xu C, Xu Y, Yang Z, Zhang Y, Tao D (2023) A survey on vision transformer. IEEE Trans Pattern Anal Mach Intell 45(1):87–110. https://doi.org/10.1109/TPAMI.2022.3152247

Harley AW, Ufkes A, Derpanis KG Evaluation of deep convolutional nets for document image classification and retrieval. In: International conference on document analysis and recognition (ICDAR)

He K, Gkioxari G, Dollár P, Girshick R (2017) Mask r-cnn. In: Proceedings of the IEEE international conference on computer vision, pp 2961–2969

Hewlett D, Lacoste A, Jones L, Polosukhin I, Fandrianto A, Han J, Kelcey M, Berthelot D (2016) Wikireading: a novel large-scale language understanding task over Wikipedia. In: Proceedings of the 54th annual meeting of the association for computational linguistics, ACL 2016, August 7–12, Berlin volume 1: long papers, the association for computer linguistics, 2016. https://doi.org/10.18653/v1/p16-1145

He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: 2016 IEEE conference on computer vision and pattern recognition, CVPR 2016, Las Vegas, June 27–30, 2016, IEEE Computer Society, pp 770–778. https://doi.org/10.1109/CVPR.2016.90

Hong T, Kim D, Ji M, Hwang W, Nam D, Park S (2022) BROS: a pre-trained language model focusing on text and layout for better key information extraction from documents. In: Thirty-sixth AAAI conference on artificial intelligence, AAAI 2022, thirty-fourth conference on innovative applications of artificial intelligence, IAAI 2022, the twelveth symposium on educational advances in artificial intelligence, EAAI 2022 virtual event, February 22-March 1, 2022, AAAI Press, pp 10767–10775. https://doi.org/10.1609/aaai.v36i10.21322

Hsiao C-W, Sun C, Sun M, Chen H-T (2019) Flat2layout: flat representation for estimating layout of general room types. arXiv:1905.12571

Huang Z, Chen K, He J, Bai X, Karatzas D, Lu S, Jawahar C. V (2019) Icdar2019 competition on scanned receipt ocr and information extraction, 2019 international conference on document analysis and recognition (ICDAR). https://doi.org/10.1109/ICDAR.2019.00244

Huang Y, Lv T, Cui L, Lu Y, Wei F (2022) Layoutlmv3: Pre-training for document ai with unified text and image masking. In: Proceedings of the 30th ACM international conference on multimedia, pp 4083–4091

Igorevna AE, Bulatovich BK, Petrovich ND, Olegovna PO, Igorevich SB, Anatolevich AnatolSO (2022) Document image analysis and recognition: a survey, computer optics. Comput Opt 46(4):567–589

Irie K, Zeyer A, Schlüter R, Ney H (2019) Language modeling with deep transformers, arXiv preprint arXiv:1905.04226

Jaume G, Ekenel HK, Thiran J (2019) FUNSD: a dataset for form understanding in noisy scanned documents. In: 2nd international workshop on open services and tools for document analysis, OST@ICDAR 2019, Sydney, September 22–25, IEEE, 2019, pp. 1–6. https://doi.org/10.1109/ICDARW.2019.10029

Kim G, Hong T, Yim M, Nam J, Park J, Yim J, Hwang W, Yun S, Han D, Park S (2022) Ocr-free document understanding transformer. arXiv:2111.15664

Lee C.-Y, Li C.-L, Dozat T, Perot V, Su G, Hua N, Ainslie J, Wang R, Fujii Y, Pfister T (2022) Formnet: structural encoding beyond sequential modeling in form document information extraction. arXiv:2203.08411

Lewis DD, Agam G, Argamon SE, Frieder O, Grossman DA, Heard J (2011) Cdip dataset, https://ir.nist.gov/cdip/

Lewis DD, Agam G, Argamon SE, Frieder O, Grossman DA, Heard J (2021) The iit cdip test collection, https://ir.nist.gov/cdip/README.txt

Lewis DD, Agam G, Argamon S, Frieder O, Grossman D. A, Heard J (2006) Building a test collection for complex document information processing. In: EN Efthimiadis, ST Dumais, D Hawking, K Järvelin (eds.), SIGIR 2006: proceedings of the 29th annual international ACM Sigir conference on research and development in information retrieval, Seattle, August 6–11, 2006, ACM, pp 665–666. https://doi.org/10.1145/1148170.1148307

Li S, Ma X, Pan S, Hu J, Shi L, Wang Q (2021) Vtlayout: fusion of visual and text features for document layout analysis. In: Pham DN, Theeramunkong T, Governatori G, Liu F (eds) PRICAI 2021: trends in artificial intelligence. Springer International Publishing, Cham, pp 308–322

Liang J, Doermann D, Li H (2005) Camera-based analysis of text and documents: a survey. Int J Doc Anal Recogn (IJDAR) 7:84–104

Li C, Bi B, Yan M, Wang W, Huang S, Huang F, Si L (2021) Structural LM: structural pre-training for form understanding. In: Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers), association for computational linguistics, pp 6309–6318. https://doi.org/10.18653/v1/2021.acl-long.493. https://aclanthology.org/2021.acl-long.493

Li P, Gu J, Kuen J, Morariu V. I, Zhao H, Jain R, Manjunatha V, Liu H (2021) Selfdoc: self-supervised document representation learning. arXiv:2106.03331

Li Y, Liang F, Zhao L, Cui Y, Ouyang W, Shao J, Yu F, Yan J (2021) Supervision exists everywhere: a data efficient contrastive language-image pre-training paradigm, arXiv preprint arXiv:2110.05208

Li Q, Li Z, Cai X, Du B, Zhao H (2023) Enhancing visually-rich document understanding via layout structure modeling. In: Proceedings of the 31st ACM international conference on multimedia, pp 4513–4523

Lin T, Dollár P, Girshick RB, He K, Hariharan B, Belongie SJ (2017) Feature pyramid networks for object detection. In: 2017 IEEE conference on computer vision and pattern recognition, CVPR 2017, Honolulu, 21–26, 2017, IEEE Computer Society, pp 936–944. https://doi.org/10.1109/CVPR.2017.106

Li Y, Qian Y, Yu Y, Qin X, Zhang C, Liu Y, Yao K, Han J, Liu J, Ding E (2021) Structext: structured text understanding with multi-modal transformers. arXiv:2108.02923

Liu Y, Yang B, Liu Q, Li Z, Ma Z, Zhang S, Bai X (2024) Textmonkey: an ocr-free large multimodal model for understanding document, arXiv preprint arXiv:2403.04473

Li B, Wang J, Zhao M, Zhou S (2022) Two-stage multimodality fusion for high-performance text-based visual question answering. In: Proceedings of the Asian conference on computer vision, pp 4143–4159

Lu J, Batra D, Parikh D, Lee S (2019) Vilbert: pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. Adv Neural Inform Process Syst 32:89

Luo C, Shen Y, Zhu Z, Zheng Q, Yu Z, Yao C (2024) Layoutllm: layout instruction tuning with large language models for document understanding. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 15630–15640

Mathew M, Karatzas D, Jawahar C (2021) Docvqa: a dataset for vqa on document images, https://www.docvqa.org/datasets

Mathew M, Karatzas D, Jawahar C (2021) Docvqa: a dataset for vqa on document images. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 2200–2209

Michael J, Labahn R, Grüning T, Zöllner J (2019) Evaluating sequence-to-sequence models for handwritten text recognition. arXiv:1903.07377

Microsoft, What is optical character recognition?, https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview-ocr#read-api (2022)

Nogueira R, Jiang Z, Cho K, Lin J (2020) Navigation-based candidate expansion and pretrained language models for citation recommendation. Scientometrics 125:3001–3016

Oral B, Eryiğit G (2022) Fusion of visual representations for multimodal information extraction from unstructured transactional documents. Int J Doc Anal Recognit (IJDAR) 25(3):187–205

Park S, Shin S, Lee B, Lee J, Surh J, Seo M, Lee H (2019) Cord: A consolidated receipt dataset for post-ocr parsing. In: Workshop on document intelligence at NeurIPS

Park S, Shin S, Lee B, Lee J, Surh J, Seo M, Lee H (2019) Cord: a consolidated receipt dataset for post-ocr parsing. In: Workshop on document intelligence at NeurIPS

Peng Q, Pan Y, Wang W, Luo B, Zhang Z, Huang Z, Hu T, Yin W, Chen Y, Zhang Y, et al (2022) Ernie-layout: Layout knowledge enhanced pre-training for visually-rich document understanding, arXiv preprint arXiv:2210.06155

Pham P, Nguyen LT, Pedrycz W, Vo B (2023) Deep learning, graph-based text representation and classification: a survey, perspectives and challenges. Artif Intell Rev 56(6):4893–4927

Powalski R, Borchmann Ł, Jurkiewicz D, Dwojak T, Pietruszka M, Pałka G (2021) Going full-tilt boogie on document understanding with text-image-layout transformer. In: Document analysis and recognition–ICDAR 2021: 16th international conference, Lausanne, September 5–10, 2021, Proceedings, Part II 16, Springer, pp 732–747

Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, Zhou Y, Li W, Liu PJ (2020) Exploring the limits of transfer learning with a unified text-to-text transformer. J Mach Learn Res 21(1):5485–5551

Ren S, He K, Girshick R, Sun J (2015) Faster r-cnn: towards real-time object detection with region proposal networks. Adv Neural Inform Process Syst 28:9

Schmidt H, Butter K, Rider C (2002) Building digital tobacco industry document libraries at the university of California, San Francisco library/center for knowledge management, D Lib Mag. 8 (9) . https://doi.org/10.1045/september2002-schmidt

Shigarov A (2023) Table understanding: problem overview. Wiley Interdiscip Rev 13(1):e1482

Shi Y, Kim M, Chae Y (2023) Multi-scale cell-based layout representation for document understanding. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision (WACV), pp 3670–3679

Shi D, Liu S, Du J, Zhu H (2023) Layoutgcn: a lightweight architecture for visually rich document understanding. In: International conference on document analysis and recognition, Springer, pp 149–165

Shinyama Y, et al (2019) Pdfminer, https://github.com/euske/pdfminer

Subramani N, Matton A, Greaves M, Lam A (2020) A survey of deep learning approaches for ocr and document understanding, arXiv preprint arXiv:2011.13534

Tanaka R, Iki T, Nishida K, Saito K, Suzuki J (2024) Instructor: a dataset for zero-shot generalization of visual document understanding with instructions. In: Proceedings of the AAAI conference on artificial intelligence, vol 38, pp 19071–19079

Tang Z, Yang Z, Wang G, Fang Y, Liu Y, Zhu C, Zeng M, Zhang C, Bansal M (2023) Unifying vision, text, and layout for universal document processing. arXiv:2212.02623

UCSF library, industry documents library, https://www.industrydocuments.ucsf.edu/

Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł (2017) Attention is all you need. Adv Neural Inform Process Syst 30:5998–6008

Voutharoja BP, Qu L, Shiri F (2023) Language independent neuro-symbolic semantic parsing for form understanding. arXiv:2305.04460

Vu HM, Nguyen DT (2020) Revising FUNSD dataset for key-value detection in document images, CoRR abs/2010.05322. arXiv:2010.05322

Wang J, Gou C, Wu Q, Feng H, Han J, Ding E, Wang J (2022) Rtformer: efficient design for real-time semantic segmentation with transformer. Adv Neural Inform Process Syst 35:7423–7436

Wang J, Jin L, Ding K (2022) Lilt: a simple yet effective language-independent layout transformer for structured document understanding. arXiv:2202.13669

Wang Y, Lipka N, Rossi RA, Siu A, Zhang R, Derr T (2024) Knowledge graph prompting for multi-document question answering. In: Proceedings of the AAAI conference on artificial intelligence, vol 38, pp 19206–19214

Wang C, Li M, Smola A. J (2019) Language models with transformers. arXiv:1904.09408

Wang X, Wang X, Jiang B, Tang J, Luo B (2023) Mutualformer: multi-modality representation learning via cross-diffusion attention arXiv:2112.01177

Wu Y, Schuster M, Chen Z, Le Q. V, Norouzi M, Macherey W, Krikun M, Cao Y, Gao Q, Macherey K, Klingner J, Shah A, Johnson M, Liu X, Kaiser Łukasz, Gouws S, Kato Y, Kudo T, Kazawa H, Stevens K, Kurian G, Patil N, Wang W, Young C, Smith J, Riesa J, Rudnick A, Vinyals O, Corrado G, Hughes M, Dean J (2016) Google's neural machine translation system: bridging the gap between human and machine translation. arXiv:1609.08144

Wu H, Xiao B, Codella N, Liu M, Dai X, Yuan L, Zhang L (2021) Cvt: introducing convolutions to vision transformers. arXiv:2103.15808

Xie S, Girshick R. B, Dollár P, Tu Z, He K (2017) Aggregated residual transformations for deep neural networks. In: 2017 IEEE conference on computer vision and pattern recognition, CVPR 2017, Honolulu, July 21–26, 2017, IEEE Computer Society, pp 5987–5995. https://doi.org/10.1109/CVPR.2017.634

Xu Y, Li M, Cui L, Huang S, Wei F, Zhou M (2020) Layoutlm: pre-training of text and layout for document image understanding. In: R Gupta, Y Liu, J Tang, BA Prakash (eds.), KDD '20: the 26th ACM SIGKDD conference on knowledge discovery and data mining, virtual event, August 23–27, ACM, 2020, pp 1192–1200. https://doi.org/10.1145/3394486.3403172

Xu Y, Lv T, Cui L, Wang G, Lu Y, Florencio D, Zhang C, Wei F (2021) Layoutxlm: Multimodal pre-training for multilingual visually-rich document understanding. arXiv:2104.08836

Xu Y, Xu Y, Lv T, Cui L, Wei F, Wang G, Lu Y, Florêncio DAF, Zhang C, Che W, Zhang M, Zhou L (2021) Layoutlmv2: multi-modal pre-training for visually-rich document understanding. In: ACL/IJCNLP

Xu P, Zhu X, Clifton DA (2023) Multimodal learning with transformers: a survey. arXiv:2206.06488

Yu Y, Li Y, Zhang C, Zhang X, Guo Z, Qin X, Yao K, Han J, Ding E, Wang J (2023) Structextv2: masked visual-textual prediction for document image pre-training, arXiv preprint arXiv:2303.00289

Yu W, Lu N, Qi X, Gong P, Xiao R (2021) PICK: processing key information extraction from documents using improved graph learning-convolutional networks. In: 25th international conference on pattern recognition, ICPR 2020, Virtual Event/Milan, January 10–15, IEEE, 2020, pp 4363–4370. https://doi.org/10.1109/ICPR48806.2021.9412927

Zhai M, Li Y, Qin X, Yi C, Xie Q, Zhang C, Yao K, Wu Y, Jia Y (2023) Fast-structext: An efficient hourglass transformer with modality-guided dynamic token merge for document understanding, arXiv preprint arXiv:2305.11392

Zhang P, Xu Y, Cheng Z, Pu S, Lu J, Qiao L, Niu Y, Wu F (2020) TRIE: end-to-end text reading and information extraction for document understanding. In: CW Chen, R Cucchiara, X Hua, G Qi, E Ricci, Z Zhang, R Zimmermann (eds.), MM '20: The 28th ACM international conference on multimedia, virtual event/Seattle, October 12–16, ACM, 2020, pp 1413–1422. https://doi.org/10.1145/3394171.3413900

Zheng G, Zhou X, Li X, Qi Z, Shan Y, Li X (2023) Layoutdiffusion: controllable diffusion model for layout-to-image generation. arXiv:2303.17189

Zhong X, Tang J, Jimeno-Yepes A (2019) Publaynet: largest dataset ever for document layout analysis. In: 2019 international conference on document analysis and recognition, ICDAR 2019, Sydney, September 20–25, 2019, IEEE, pp. 1015–1022. https://doi.org/10.1109/ICDAR.2019.00166